arXiv:2604.22366v2 [math.OC] 2 Sep 2026

Statistical Estimation of Monge Transport Maps via Brenier Potentials

Elsa Cazelles
CNRS, IRIT, Université de Toulouse

Edouard Pauwels
Toulouse School of Economics
Université Toulouse Capitole

Léo Portales
IRIT, TSE, INP Toulouse

September 3, 2026

Abstract

We introduce and analyze a statistical estimator for Monge transport maps: solutions to the quadratic optimal transport problem in the Euclidean space. For absolutely continuous source measures, this map is uniquely defined as the gradient of a convex function, a result known as Brenier's theorem. Without absolute continuity, the problem is relaxed, maps are replaced by coupling measures, and optimal couplings are supported on the subdifferential of a convex function, called a Brenier potential. This potential is the basis for our transport map statistical estimator, for measures known only through finite samples. We construct an explicit optimal Brenier potential for the empirical problem, with a simple closed form expression based on dual solutions. Our transport map estimator is then defined as a suitable subdifferential selection, and can be extended to $\mathbb{R}^d$. We exhibit convergence rates for this estimator based on a new error bound for the quadratic optimal transport problem. In the semi-discrete setting, where the target measure is finitely supported, our estimator enjoys sharper convergence rates. Our methodology does not rely on smoothness or continuity of the target Monge transport map and requires no computation beyond primal-dual solutions of the empirical finite dimensional problem. Finally, using similar proof techniques, we provide novel convergence rate for empirical couplings.

1 Introduction

Optimal transport, as initially formulated by Monge, seeks a map which "pushes-forward" a source probability measure μ onto a target probability measure ν while minimizing the quadratic cost. The classical Monge problem is written over $\mathbb{R}^d$ as follows, over the class of measurable mappings T

$$\inf_{\substack{T\colon \mathbb{R}^d\to\mathbb{R}^d \\ T_\sharp\mu=\nu}} \int_{\mathbb{R}^d} \frac{1}{2}\|x-T(x)\|^2 d\mu(x), \tag{1.1}$$

where $\sharp$ denotes the push forward operation. The constraint can be equivalently formulated as $\int f(y)d\nu(y) = \int f(T(x))d\mu(x)$ for all measurable f on $\mathbb{R}^d$. A solution to this problem, when it exists, will be called a *Monge map*.

Optimal transport is one of the key mathematical problems of the late 20-th century, both as an analytical tool and as a mathematical modeling framework with many applications [Villani, 2008, Santambrogio, 2015,

Peyré and Cuturi, 2019]. We limit ourselves to the quadratic case, as in (1.1). Our question of interest is the statistical estimation of T for source and target μ, ν known only through independent samples, or equivalently, based on discrete empirical measures $\hat{\mu}_n, \hat{\nu}_m$. We introduce a general estimator for T, and provide finite sample guaranties under mild assumptions on problem data: compact support of the source and target measures, and absolute continuity of the source measure.

Monge problem admits a (semi)dual formulation over the set of convex functions on $\mathbb{R}^d$ as follows,

$$\inf_{\varphi \in \Gamma_0(\mathbb{R}^d)} \int_{\mathbb{R}^d} \varphi(x) d\mu(x) + \int_{\mathbb{R}^d} \varphi^*(y) d\nu(y), \tag{1.2}$$

where $\Gamma_0(\mathbb{R}^d)$ denotes the set of convex lower semicontinuous proper functions over $\mathbb{R}^d$ and $\varphi^* \colon y \mapsto \sup_{x \in \mathbb{R}^d} \langle x, y \rangle - \varphi(x)$ is the Legendre-Fenchel conjugate. A celebrated result of Brenier [Brenier, 1991] ensures that for an absolutely continous source measure μ, the solution to (1.1) exists, is unique and is the (sub)gradient of a convex function $\varphi \in \Gamma_0(\mathbb{R}^d)$. This function φ turns out to be an optimal solution to problem (1.2) [Santambrogio, 2015]. Solutions to (1.2) will be called Brenier potentials.

These elements are well known [Villani, 2008, Villani, 2021, Santambrogio, 2015], and we provide a complete description in Section 2. We also expand on the discrete optimal transport problem, between finitely supported measures, with the corresponding finite dimensional linear programming formulation [Peyré and Cuturi, 2019].

1.1 Overview of the main contributions

An explicit Brenier potential For discrete measures μ_n and ν_m, supported on n and m points respectively, we show that the dual problem (1.2) admits an explicit solution which takes the form of the maximum of m affine functions. The coefficients of these affine functions are directly deduced from the support points of the target ν_m and the dual solution of the discrete optimal transport linear program, see Section 3.1 and in particular Definition 3.1. This Brenier potential has basically the same computational cost as computing the discrete optimal transport problem, as most solvers provide primal-dual solutions. Its structure is reminiscent of the Laguerre tesselation occurring in semi-discrete optimal transport [Aurenhammer et al., 1998], but in a discrete-discrete setting. This construction serves as a basis for our statistical estimator.

Finite sample guaranties In a statistical setting, given population source (which is absolutely continuous) and target probability measures μ, ν, and empirical measures $\hat{\mu}_n$, $\hat{\nu}_m$ supported on n and m points respectively, we have the following under mild assumptions.

- As known, the population problem (1.1) for (μ, ν) admits a solution $T = \nabla\bar{\varphi}$, where $\bar{\varphi}$ solves (1.2).
- For the discrete random problem between $\hat{\mu}_n$ and $\hat{\nu}_m$, we have an explicit Brenier map estimator $\hat{\phi}$, solution to (1.2) as described in the previous paragraph.

It is therefore natural to consider $\nabla\hat{\phi}$ as a statistical estimator for $\nabla\bar{\varphi}$. This estimator enjoys computational simplicity (closed form expression), without any parameter to estimate or tune.

The main statistical question to be addressed is that of the convergence of $\nabla\hat{\phi}$ as the sample sizes n and m increase. Noting that the convex subdifferential (see [Rockafellar, 1970]) satisfies $\partial\hat{\phi}(x) = \{\nabla\hat{\phi}(x)\}$ for almost all x, it is possible to invoke the main results of [Segers, 2022] to obtain that, almost surely, $\partial\hat{\phi}$ converges graphically to $\partial\bar{\varphi}$, restricted to the interior of the support of μ. This is a qualitative result, illustrating the fact that $\nabla\hat{\phi}$ is a relevant estimator. However, the contribution of [Segers, 2022] remains

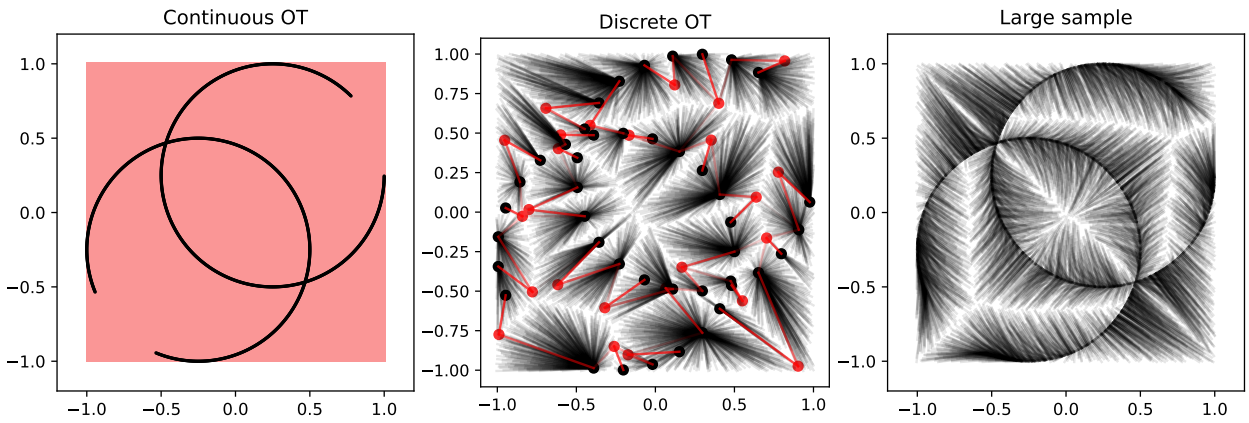

Figure 1: Visualization of the main contribution in the plane. *Left:* Population problem, the goal is to send the uniform red source density on the square to the black singular target measure. *Center:* Discrete, sampled, version of the problem with empirical measures. Dots represent sample from source (red) and target (black). The red lines represent the discrete optimal transport solutions (red mass is split as there are more black points). The black lines represent the Monge map estimator, computed from dual optimal solutions of the discrete problem. It is piecewise constant, defined on the whole plane, and each black line represents the association between a new source input and its image. *Right:* For large samples, the estimator converges to the population Monge map, illustrating our finite sample L^1 bound. Note that, in this case, the population Monge Map is not continuous and our results apply to this setting. Discrete solutions are computed using POT [Flamary et al., 2021].

qualitative and the underlying notion of convergence is difficult to explicit in functional analysis terms. We considerably extend this result for our specific estimator and provide a finite sample upper bound on the quantity $\|\nabla\hat{\phi} - \nabla\bar{\varphi}\|_{L^1(\mu)}$. This upper bound holds under mild assumptions: compact support of μ, ν, and upper bounded density of μ. In particular we make no assumption regarding the convexity of the support of μ or any lower bound on its density, and the population Monge map $\nabla\bar{\varphi}$ may be discontinuous. This is our main statistical result, it is given in Section 3.2. A visualization of the estimator and its convergence is given in Figure 1.

A dual error bound The key analytical element which allows to devise our finite sample estimate is an error bound for Problem (1.2). We show in Section 4 that for compactly supported probability measures μ, ν, the source μ with an upper bounded density and $\bar{\varphi}$ solution to (1.2), there is $c > 0$ such that for any convex function ϕ in an appropriate subclass of $\Gamma_0(\mathbb{R}^d)$, it holds that

$$c\|\nabla\phi - \nabla\bar{\varphi}\|^4_{L^1(\mu)} \leq \int \phi - \bar{\varphi} d\mu + \int \phi^* - \bar{\varphi}^* d\nu. \tag{1.3}$$

Choosing $\phi = \hat{\phi}$, our Brenier map estimator from empirical measures $\hat{\mu}_n, \hat{\nu}_m$, the right hand side can be further upper bounded by $W_1(\mu, \hat{\mu}_n) + W_1(\nu, \hat{\nu}_m)$ for which finite sample upper bounds exist in expectation in [Fournier and Guillin, 2015], see [Fournier, 2023] for explicit multiplicative constants. This results in statistical rates roughly of order $O(n^{-\frac{1}{4d}} + m^{-\frac{1}{4d}})$. We do not know if these rates could be improved for the proposed estimator, or for other estimators. The closest lower bound is $n^{-\frac{1}{d}}$, but under considerably more restrictive conditions on the measures, see Section 3.3.

The error bound is of independent interest for quadratic optimal transport beyond statistical estimation. It relates for example to stability properties for Monge maps [Gigli, 2011] with applications in machine learning [Mérigot et al., 2020]. The proof combines Carlier's remainder for Fenchel-Young inequality [Carlier, 2023] with the covering number estimate from [Carlier et al., 2025], which we use as a measure of discontinuity for gradients of Lipschitz convex functions.

Extensions The proposed framework and techniques are extended to provide refined statistical bounds for the semi-discrete optimal transport problem in Section 3.4. In Section 3.5, quantitative convergence guaranties of empirical optimal couplings toward optimal couplings are given, based on quantitative stability bounds in Section 5. The consistency of empirical coupling is due to [Segers, 2022].

1.2 Related work

Statistical optimal transport Research on the topic has considerably expanded in the last decade. This includes many variations: semi-discrete, entropic regularization, convergence of the optimal transport cost, of the optimal map, etc. We focus on the estimation of quadratic optimal transport Monge maps. We refer to [Balakrishnan et al., 2025] for a recent review and will comment here on the specificity of our approach and the closest references.

The idea of estimating Monge maps based on the dual optimal transport formulation was explored in [Hütter and Rigollet, 2021] and expanded in [Manole et al., 2024, Divol et al., 2025]. This literature deploys the tools of non parametric statistical estimation to tackle the problem. The main drawback of this approach is that it relies on smoothness of the population transport map, which is required to be an element of a well identified function space, such as that of bi-Lipschitz diffeomorphisms. Regularity of optimal transport is a topic of notable mathematical difficulty and sufficient conditions impose considerable restrictions. The main specificity of our work with respect to this literature is that we do not directly apply nonparametric estimation to $\nabla\varphi$, but rather leverage a new error bound in (1.3). This allows to shift the regularity requirements from the Monge map $\nabla\varphi$ to the Brenier potential φ and leverage known convergence rates for empirical measures [Fournier and Guillin, 2015]. As a consequence our main result holds under explicit and versatile assumptions on the problem: compacity of the supports, absolute continuity of the source, and an upper bound on its density. A striking consequence is that the target Monge map is not even required to be continuous. The only available occurrence of statistical convergence toward possibly discontinuous Monge maps is [Pooladian et al., 2023] which is limited to the semi-discrete setting. A more detailed discussion on these aspects is given in Section 3.3.

Estimators for Monge maps A diversity of Monge map estimators have been proposed in the literature with various statistical analyses. These include M-estimators [Hütter and Rigollet, 2021, Manole et al., 2024, Divol et al., 2025, Chewi et al., 2025], entropic regularization [Seguy et al., 2018, Pooladian and Niles-Weed, 2021], kernel based estimators [Perrot et al., 2016, Gunsilius, 2022], input convex neural networks [Makkuva et al., 2020], one nearest neighbors and barycentric projection [Manole et al., 2024], convex extrapolation and smoothing [Paty et al., 2020, Hallin et al., 2021, De Lara et al., 2021], kernel sums-of-squares [Vacher et al., 2021, Muzellec et al., 2021].

Interestingly, our proposed Brenier potential has not been considered explicitly. It is a natural estimator as it is the solution of Problem (1.2) with discrete measures and it admits a closed form solution based on the optimal transport linear program dual solutions. As discussed in Lemma 3.4, our estimator corresponds to the limit as entropic regularization vanishes of the estimator proposed in [Seguy et al., 2018] and analyzed

in [Pooladian and Niles-Weed, 2021, Pooladian et al., 2023]. Finally, [Vacher and Vialard, 2022] explores the distance between gradients of convex potentials and the dual loss, under smoothness condition on the source density, in a statistical setting.

Stability of optimal transport maps The error bound in (1.3) translates into a Hölder stability bound for Monge maps. This topic is of interest more broadly for optimal transport [Gigli, 2011, Mérigot et al., 2020, Li and Nochetto, 2021, Delalande and Mérigot, 2023, Letrouit and Mérigot, 2024]. Our error bound directly implies a similar stability bound, allowing for variations of both source and target probability measures. After finalizing the first version of this manuscript, we became aware of the independent work [Mérigot, 2026], providing improved exponents for this problem using techniques similar to ours, see also the discussion in the conclusion. These exponents turn out to be optimal as shown by [Letrouit, 2025]. This work also provides an error bound similar to (1.3). A more precise discussion of the relation to existing art is given in Section 4.4 and in the conclusion.

Semi-discrete optimal transport Our estimator is strongly related to Laguerre tesselation, adapted to a discrete setting, a notion which has very strong ties with semi-discrete optimal transport [Aurenhammer, 1987]. Similar piecewise linear estimators are studied in convex geometry [Gu et al., 2013], in relation to optimal transport and its applications [Lei et al., 2019].

Finally, we specify our results to the semi-discrete setting, that is for ν a discrete probability measure, showing that it is possible to remove the exponential dependency in the ambient dimension d. This is in line with existing statistical results on this topic [Pooladian et al., 2023, Sadhu et al., 2024]. Statistical aspects for the semi-discrete setting are further discussed in Section 3.4.

1.3 Notation

Throughout the paper $\|.\|$ will denote the standard Euclidean norm. For any $R > 0$, $B_R := B_d(0, R)$ and $\bar{B}_R := \bar{B}_d(0, R)$ will denote respectively the d-dimensional Euclidean open and closed ball centered in 0 with radius R. For any function f and measure μ, we denote $\mathrm{graph}(f)$ the graph of f and $\operatorname{supp}\mu$ the support of μ. For any function $f : \mathbb{R}^d \to \mathbb{R}$ we will denote f^* its Fenchel-Legendre transform, that is, for all $y \in \mathbb{R}^d$: $f^*(y) = \sup_{x\in\mathbb{R}^d}\langle x, y\rangle - f(x)$. For any set of points $\{y_i\}_{i\in I}$ we denote $\mathrm{conv}\{y_i,\ i \in I\}$ its convex hull. We also define $\Gamma_0(\mathbb{R}^d)$, the classical set of proper lower semi-continuous convex functions on $\mathbb{R}^d$, that is:

$$\Gamma_0(\mathbb{R}^d) = \left\{\phi : \mathbb{R}^d \longrightarrow \mathbb{R} \cup \{+\infty\},\ \phi \text{ is convex, l.s.c, proper } \right\}.$$

We call the domain of a function f the set of points where f is finite. For any set C, we denote $\delta(.|C)$ the indicator function which values 0 at C and $+\infty$ elsewhere. For measure subsets, we introduce the following notation. For any subset X of $\mathbb{R}^d$, we denote $\mathcal{M}_1(X)$ the set of probability measures with support included in X. Finally for any $(\mu, \nu) \in \mathcal{M}_1(X) \times \mathcal{M}_1(Y)$ we will denote the set of probability measures with respective marginals μ and ν as follows

$$\Pi(\mu, \nu) = \big\{\pi \in \mathcal{M}_1(X\times Y) | \pi(A\times Y) = \mu(A), \pi(X\times B) = \nu(B), \forall A \subset X, B \subset Y \text{measurable subsets}\big\}.$$

For any mapping $f : \mathbb{R}^d \to \mathbb{R}^d$, $f_\sharp\mu$ denotes the push-forward of measure μ by f, that is, for any measurable subset $B \subset \mathbb{R}^d$, $f_\sharp\mu(B) = \mu(f^{-1}(B))$.

2 Preliminaries on optimal transport

We briefly recall basic concepts in optimal transport as well as important notions for the paper.

2.1 Primal and dual problems

The problem in (1.1) does not always have a solution. The Kantorovich relaxation replaces maps by probability couplings with marginals μ and ν (*i.e* elements of $\Pi(\mu,\nu)$), as follows

$$\inf_{\gamma\in\Pi(\mu,\nu)} \int_{\mathbb{R}^d\times\mathbb{R}^d} \frac{1}{2}\|x-y\|^2 d\gamma(x,y). \tag{2.1}$$

Duality is typically expressed in terms of c-concave functions [Santambrogio, 2015, Section 1.6.1]. Focusing on the quadratic cost allows us to adopt a classical alternative: expressing duality with usual convexity notions. Both approaches are equivalent up to a change of variables [Santambrogio, 2015, Proposition 1.21]. Problem (2.1) admits the following dual

$$\sup_{\varphi\in\Gamma_0(\mathbb{R}^d)} \int_{\mathbb{R}^d} \left(\frac{\|x\|^2}{2} - \varphi(x)\right) d\mu(x) + \int_{\mathbb{R}^d} \left(\frac{\|y\|^2}{2} - \varphi^*(y)\right) d\nu(y). \tag{2.2}$$

This formulation is sometimes called "semi-dual", as it has a single decision variable while there are two marginal constraints in the primal. When μ is absolutely continuous, Brenier's Theorem tells us that the optimal Monge map T in (1.1) is almost-everywhere the gradient of $\varphi \in \Gamma_0(\mathbb{R}^d)$, an optimal solution to (2.2), called a *Brenier potential*. In the absence of continuity when the measures are compactly supported, optimal couplings $\gamma \in \Pi(\mu,\nu)$ exist for the Kantorovich problem (2.1). Moreover such solution γ is supported on the graph of the subdifferential of φ, *i.e.* $\operatorname{supp}\gamma \subset \operatorname{graph}(\partial\varphi)$, a solution to Problem (2.2). These are classical facts [Villani, 2008, Villani, 2021, Santambrogio, 2015] which we summarize in the following theorem. We provide a proof for the latter for completeness in Section A.2

Theorem 2.1 (Duality for quadratic optimal transport)**.** *Let $\mu,\ \nu$ be compactly supported probability measures. Consider the following problems*

$$\inf_{\gamma\in\Pi(\mu,\nu)} \int_{\mathbb{R}^d\times\mathbb{R}^d} \frac{1}{2}\|x-y\|^2 d\gamma(x,y) \tag{PP}$$

and

$$\int_{\mathbb{R}^d} \frac{\|x\|^2}{2} d\mu(x) + \int_{\mathbb{R}^d} \frac{\|y\|^2}{2} d\nu(y) - \inf_{\varphi\in\Gamma_0(\mathbb{R}^d)} \int_{\mathbb{R}^d} \varphi(x) d\mu(x) + \int_{\mathbb{R}^d} \varphi^*(y) d\nu(y). \tag{DP}$$

We have (PP) $=$ (DP) *and optimal solutions to* (PP) *and* (DP) *exist. Moreover the following are equivalent, for any $\gamma \in \Pi(\mu,\nu)$ and $\varphi \in \Gamma_0(\mathbb{R}^d)$*

1. *γ is optimal for* (PP) *and φ is optimal for* (DP)*.*
2. *$\int \frac{1}{2}\|x-y\|^2 d\gamma(x,y) = \int \left(\frac{\|x\|^2}{2} - \varphi(x)\right) d\mu(x) + \int \left(\frac{\|y\|^2}{2} - \varphi^*(y)\right) d\nu(y)$*
3. *$\operatorname{supp}\gamma \subset \operatorname{graph}(\partial\varphi)$.*

Furthermore, φ may be restricted to be R_ν-Lipschitz with $R_\nu = \max_{y\in\operatorname{supp}\nu}\|y\|$. Finally, if μ is absolutely continuous, the solution to (PP) *is unique and defined μ-almost everywhere as $\gamma = (Id, \nabla\varphi)_\sharp\mu$, and $T = \nabla\varphi$ is solution to Problem* (1.1)*.*

In this article we will rely on an alternative formulation of this theorem, given below as a lemma.

Lemma 2.2 (Local Lipschitz continuity of Brenier potentials and their conjugate)**.** *Let μ, ν be as in Lemma 2.1 with compact support respectively included in B_{R_μ} and B_{R_ν}, then optimal Brenier potentials in* (DP) *may be chosen in*

$$\mathcal{C}_{R_\mu,R_\nu} = \left\{ \varphi \in \Gamma_0(\mathbb{R}^d) \;\middle|\; \varphi \text{ } R_\nu\text{-Lipschitz on } B_{R_\mu},\ \varphi^* \text{ } R_\mu\text{-Lipschitz on } B_{R_\nu} \right\}.$$

In fact, for any φ_0 optimal in Lemma 2.1, there exists $\varphi_1 \in \mathcal{C}_{R_\mu,R_\nu}$ such that $\varphi_1 = \varphi_0$ on $\operatorname{supp}\mu$ and $\varphi_1^* = \varphi_0^*$ on $\operatorname{supp}\nu$.

Proof. Pick a R_ν Lipschitz optimal potential φ_0 in Lemma 2.1 and γ an optimal coupling. We have $\operatorname{supp}\gamma \subset \operatorname{graph}\partial\varphi$ and $\operatorname{supp}\mu \subset B_{R_\mu}$. Furthermore $\operatorname{supp}\nu \subset \partial\varphi(B_{R_\mu})$ using Lemma A.6 and noticing that $\{y \in \mathbb{R}^d,\ \exists x \in \operatorname{supp}\mu,\ (x,y) \in \operatorname{supp}\gamma\} \subset \partial\varphi(B_{R_\mu})$ using the third point of Theorem 2.1.

Lemma A.1 ensures that there is $\varphi_1 \in \mathcal{C}_{R_\mu,R_\nu}$ such that $\varphi_1 = \varphi_0$ on $\operatorname{supp}\ \mu$ and $\varphi_1^* = \varphi_0^*$ on $\operatorname{supp}\ \nu$. We deduce that $\int \varphi_1 d\mu = \int \varphi_0 d\mu$ and $\int \varphi_1^* d\nu = \int \varphi_0^* d\nu$ so that φ_1 is also optimal by point 2 of Theorem 2.1 □

Remark 2.3. *Any $\varphi \in \Gamma_0(\mathbb{R}^d)$ Lipschitz over B_{R_μ} is also Lipschitz over $\overline{B}_{R_\mu}$. This is because Lipschitz continuity allows to extend φ from the interior to the boundary, and this extension corresponds to the actual value of φ by lower semi-continuity.*

Lemma 2.2 is a regularity result on the solution of problem (DP). Its specificity lies in the fact that both φ and φ^* are considered. Note that both cannot be globally Lipschitz at the same time. Let us mention that Lipschitz continuity of Kantorovich potentials was considered in [McCann, 2001], and that fine regularity is a crucial topic in optimal transport, starting with the work of Caffarelli [Caffarelli, 1992a, Caffarelli, 1992b, Caffarelli, 1996].

2.2 Optimal transport between finitely supported measures

Our basic framework for statistical estimation is for finitely supported source and target measures. Throughout this section, we consider the special case when $\hat{\mu} = \sum_{i=1}^n a_i \delta_{x_i}$, $\hat{\nu} = \sum_{j=1}^m b_j \delta_{y_j}$, where $a_i > 0$ for $i = 1, \dots, n$, $b_j > 0$ for $j = 1, \dots, m$, such that $\sum_{i=1}^n a_i = \sum_{j=1}^m b_j = 1$, and $x_1, \dots, x_n$, $y_1, \dots, y_m$ are atoms in $\mathbb{R}^d$. The optimal transport problem between such discrete measures falls within the scope of Lemma 2.1, but can also be finitely represented, resulting in a linear program. This has been extensively studied, see for example [Peyré and Cuturi, 2019] for a recent account of computational aspects.

The primal problem (PP) is written as a finite dimensional linear program over matrices [Peyré and Cuturi, 2019, Section 2.3]:

$$\min \left\{ \langle C, P \rangle \;\middle|\; P \in \mathbb{R}_+^{n \times m},\ \sum_{j=1}^m P_{i,j} = a_i,\ \sum_{i=1}^n P_{i,j} = b_j,\ \forall i = 1, \dots, n,\ j = 1, \dots, m \right\}, \qquad \text{(PP-discr)}$$

where $C \in \mathbb{R}^{n \times m}$ is a cost matrix defined as $C_{i,j} = \frac{1}{2}\|x_i - y_j\|^2$ for all $i = 1, \dots, n,\ j = 1, \dots, m$. This is obtained by identifying the matrix P above with a coupling γ between $\hat{\mu}$ and $\hat{\nu}$ as in Lemma 2.1, with the identification $\int f(x,y) d\gamma(x,y) = \sum_{i,j} f(x_i, y_j) P_{i,j}$ for all measurable f.

The corresponding dual linear program is given as follows [Peyré and Cuturi, 2019, Section 2.5]

$$\max \left\{ \langle f, a \rangle + \langle g, b \rangle \;\middle|\; f \in \mathbb{R}^n,\ g \in \mathbb{R}^m,\ f_i + g_j \leq C_{i,j},\ \forall i = 1, \dots, n,\ j = 1, \dots, m \right\}. \qquad \text{(DP-discr)}$$

An application of strong duality ensures that the values of (PP-discr) and (DP-discr) are the same and both are attained [Peyré and Cuturi, 2019, Proposition 2.4]. Unlike the coupling matrix P in (PP-discr), the dual variables in (DP-discr) do not have a direct interpretation in terms of convex functions as in Lemma 2.1.

3 Main results on the estimation of Brenier potentials and Brenier maps

3.1 An estimator based on discrete dual solutions

We are now in position to describe our Brenier potential estimator, which is based on the following explicit Brenier potential.

Definition 3.1 (Explicit Brenier potential)**.** *Let $\hat{\mu}, \hat{\nu}$ be two discrete measures over $\mathbb{R}^d$, such that $\hat{\mu} = \sum_{i=1}^n a_i \delta_{x_i}$, $\hat{\nu} = \sum_{j=1}^m b_j \delta_{y_j}$, where $a_i > 0$, $i = 1 \dots n$, $b_j > 0$, $j = 1, \dots, m$, $\sum_{i=1}^n a_i = \sum_{j=1}^m b_j = 1$ and $x_1, \dots, x_n$, $y_1, \dots, y_m$ atoms in $\mathbb{R}^d$. Let $g \in \mathbb{R}^m$ be an optimal solution for* (DP-discr)*, the associated Brenier potential is given by*

$$\varphi_g \colon x \in \mathbb{R}^d \mapsto \max_{k=1,\dots,m} \langle x, y_k \rangle + g_k - \frac{1}{2}\|y_k\|^2. \tag{3.1}$$

Complementary slackness for linear programming allows to verify the optimality condition from Lemma 2.1.

Lemma 3.2 (Optimality of the explicit potential)**.** *Let $\hat{\mu}, \hat{\nu}$ be as in Lemma 3.1, let P be an optimal matrix in the primal* (PP-discr) *and (f, g) be an optimal pair of dual variables in* (DP-discr)*. Then*

$$P_{i,j} > 0 \implies y_j \in \partial \varphi_g(x_i)$$

where φ_g is given in Lemma 3.1. As a consequence φ_g is optimal for (DP) *with $\mu = \hat{\mu}$, $\nu = \hat{\nu}$, and*

$$\min_{\varphi \in \Gamma_0(\mathbb{R}^d)} \int \varphi d\hat{\mu} + \int \varphi^* d\hat{\nu} = \min_{g \in \mathbb{R}^m} \int \varphi_g d\hat{\mu} + \int \varphi_g^* d\hat{\nu}.$$

Proof. The constraint in (DP-discr) writes for all $k = 1, \dots, m$, $i = 1, \dots, n$, $f_i \leq \frac{1}{2}\|x_i - y_k\|^2 - g_k$. For any $i = 1, \dots, n$, $j = 1, \dots, m$ such that $P_{i,j} > 0$, complementary slackness [Peyré and Cuturi, 2019, Proposition 3.2] implies that $f_i + g_j = \frac{1}{2}\|x_i - y_j\|^2$, and $f_i = \min_{k=1,\dots,m} \frac{1}{2}\|x_i - y_k\|^2 - g_k$ with minimizing index j. This results in the following implications.

$$\begin{aligned}
P_{i,j} > 0 &\implies \min_{k=1,\dots,m} \frac{1}{2}\|x_i - y_k\|^2 - g_k = \frac{1}{2}\|x_i - y_j\|^2 - g_j \\
&\iff \max_{k=1,\dots,m} \langle x_i, y_k \rangle - \frac{\|y_k\|^2}{2} + g_k = \langle x_i, y_j \rangle - \frac{\|y_j\|^2}{2} + g_j \\
&\iff \varphi_g(x_i) = \langle x_i, y_j \rangle - \frac{\|y_j\|^2}{2} + g_j \quad \implies \quad y_j \in \partial \varphi_g(x_i),
\end{aligned}$$

where we used that, classically (see *e.g.* [Rockafellar, 1970, Theorem 23.9]), $\partial \varphi_g(x_i) = \operatorname{conv}\{y_j,\ \varphi_g(x_i) = \langle x_i, y_j \rangle - \frac{\|y_j\|^2}{2} + g_j\}$. We identify P with an optimal coupling $\gamma \in \Pi(\hat{\mu}, \hat{\nu})$ so that $P_{i,j} > 0 \iff (x_i, y_j) \in \operatorname{supp}(\gamma)$. This allows us to use the third point of Theorem 2.1 showing that φ_g is an optimal Brenier potential for (DP). □

The potential φ_g in Lemma 3.1 is convex and globally Lipschitz. As such, its gradient is well defined Lebesgue almost everywhere, for example using Rademacher's theorem. Setting for all $x \in \mathbb{R}^d$, $I(x) = \{j = 1, \dots, m | \varphi_g(x) = \langle x, y_j \rangle + g_j - \frac{1}{2}\|y_j\|^2\}$, the (non-empty) set of active indices, we then have $\partial\varphi_g(x) = \mathrm{conv}\{y_j,\, j \in I(x)\}$ for all $x \in \mathbb{R}^d$. As the maximum of affine functions, $\partial\varphi_g$ is a singleton almost everywhere, and any selection is a transport map, extended to $\mathbb{R}^d$, for discrete measures. Note that any selection in the subdifferential is measurable. Indeed, by [Rockafellar, 1970, Theorem 25.5], it coincides with $\nabla\varphi_g$ on a set of full Lebesgue measure, and, on this set, $\nabla\varphi_g$ is continuous, hence measurable. Moreover, any selection may not coincide with $\nabla\varphi_g$ only on a Lebesgue-negligible set, hence measurable by completeness. This ensures that our estimator is μ-measurable by absolute continuity of μ with respect to the Lebesgue measure. In what follows we will consider the following simple selection.

Definition 3.3 (Transport map extension)**.** *Let $\hat{\mu}, \hat{\nu}$ be as in Lemma 3.1, and let (f, g) be optimal in* (DP-discr)*. Any selection $\hat{T}\colon \mathbb{R}^d \to \mathbb{R}^d$, such that $\hat{T}(x) \in \partial\varphi_g(x)$ for all $x \in \mathbb{R}^d$ is a transport map extension. For example,*

$$\hat{T}(x) = \frac{\sum_{j \in I(x)} y_j}{\sum_{j \in I(x)} 1}, \qquad x \in \mathbb{R}^d.$$

In the following sections, we will consider the transport map defined in Lemma 3.3 for empirical probability measures $\hat{\mu}_n, \hat{\nu}_m$, which defines our proposed estimator.

Remark 3.4. *The estimator proposed in [Seguy et al., 2018] and analyzed in [Pooladian and Niles-Weed, 2021, Pooladian et al., 2023] is based on the entropic regularized version of* (PP-discr)*, a widespread approach in numerical optimal transport [Cuturi, 2013, Peyré and Cuturi, 2019]. With entropic regularization, the dual in* (DP-discr) *remains very similar, the constraint being replaced by an exponential penalty term in the functional. Using the main result of [Cominetti and Martín, 1994], see also [Genevay et al., 2016, Weed, 2018], as the regularization penalty parameter ϵ vanishes, the primal-dual solutions of the entropic regularized problem converge to primal-dual solutions of the unregularized problem. Combining this with asymptotic properties of exponential weights, the estimator in Lemma 3.3 is exactly the limit as $\epsilon \to 0$ of the estimator of [Seguy et al., 2018, Pooladian and Niles-Weed, 2021, Pooladian et al., 2023] based on entropic regularization.*

From a practical perspective, the main advantage of the estimator in Lemma 3.1 compared to entropic regularization is that there is no regularization parameter ϵ to tune, and that it bypasses the hurdles of running a variant of the Sinkhorn algorithm for small ϵ, typically required for transport map estimation, a notorious numerical issue [Peyré and Cuturi, 2019, Chhaibi et al., 2025]. On the other hand, having one less hyperparameter to tune may be a disadvantage as a well tuned ϵ should perform at least as good as the vanishing limit.

Remark 3.5 (On the uniqueness of the estimator)**.** *As is well known in optimal transport, there can be several optimal solutions for the dual vector g in the discrete problem, potentially leading to different Monge map estimators, even when enforcing $g_m = 0$ to factor out the invariance up to additive constants.*

Uniqueness of dual potentials is of considerable interest from both a computational (e.g., reproducibility) and a statistical perspective, as it plays a role in establishing asymptotic Gaussianity of optimal transport quantities; see, e.g., [Del Barrio et al., 2024]. However, in the discrete (empirical) case, it is easy to construct examples where the configuration of the supports leads to several optimal dual variables. Even when the target measure is continuous, uniqueness of dual potentials can be difficult to establish; see [Staudt et al., 2025].

In practice, when several optimal solutions exist, the particular solution returned depends on the solver and its implementation (e.g., vertices for the simplex algorithm, a centroid of a face for interior-point methods, or a randomly selected solution for stochastic algorithms). Our theory applies to any choice of the optimal dual potentials. We are not aware of generally applicable strategies that would allow one to select a particular optimal vector g in the definition of our Monge map estimator across a wide family of solvers. We conjecture that most discrete configurations only lead to unique dual vectors (up to the usual invariance). We leave a more precise investigation for future work.

3.2 Statistical setting and convergence of the Monge map estimator

In this section, we consider two compactly supported population measures, the source having a density. In a statistical setting, they are approximated by empirical measures, obtained from i.i.d samples, as summarized in the following.

Assumption 1. *Let $R > 0$ and $M > 0$ be scalars such that μ, ν are compactly supported in B_R, and μ has density upper-bounded by M. For any positive integers n and m, consider i.i.d random samples $X_1 \dots, X_n$ and $Y_1, \dots, Y_m$ with respective distribution μ and ν. We then define the empirical probability measures $\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}$ and $\hat{\nu}_m = \frac{1}{m}\sum_{i=1}^m \delta_{Y_i}$. The Monge map estimator is $\hat{T}_{n,m}$ as in Lemma 3.3 for empirical discrete measures $\hat{\mu}_n$ and $\hat{\nu}_m$.*

Our main result relies on a novel error bound, adapted to our non-smooth estimator. Complete proof details can be found in Section 4.

Theorem 3.6 (Lemma 4.8)**.** *For any d, R, M there is C positive such that the following holds. Let μ, ν be probability measures compactly supported in B_R such that $\mu(A) \leq M\lambda(A)$ for any measurable set $A \subset \mathbb{R}^d$. Let $\bar{\varphi}$ be an R-Lipschitz Brenier potential for μ and ν, optimal for* (DP) *as in Lemma 2.1. For any $\hat{\mu}$, $\hat{\nu}$ supported on B_R with $\phi_0 \in \Gamma_0(\mathbb{R}^d)$ an associated Brenier potential which is $2R$-Lipschitz on B_{2R}, we have*

$$\int_{\mathbb{R}^d} \|\nabla\bar{\varphi}(x) - \nabla\phi_0(x)\| d\mu(x) \leq C\left(W_1(\mu,\hat{\mu}) + W_1(\nu,\hat{\nu})\right)^{\frac{1}{4}}.$$

Our main statistical result for the proposed estimator is an application of Lemma 3.6.

Theorem 3.7. *Under Assumption 1 for all $M > 0$, $R > 0$ and $d \in \mathbb{N}^*$ there exists $C > 0$ such that the following holds. Let $\bar{\varphi}$ be an optimal Brenier potential, optimal in* (DP) *for measures μ and ν. For measures $\hat{\mu}_n$ and $\hat{\nu}_m$, n, $m \in \mathbb{N}^*$. Then*

$$\mathbb{E}\left[\int_{\mathbb{R}^d} \|\nabla\bar{\varphi}(x) - \hat{T}_{n,m}(x)\| d\mu(x)\right] \leq C \begin{cases} \left(n^{-1/2} + m^{-1/2}\right)^{1/4} & \text{if } d = 1, \\ \left(n^{-1/2}\log(1+n) + m^{-1/2}\log(1+m)\right)^{1/4} & \text{if } d = 2, \\ \left(n^{-1/d} + m^{-1/d}\right)^{1/4} & \text{if } d > 2. \end{cases}$$

Proof. From Definition 3.3 there exists $g \in \mathbb{R}^m$ and a function $\varphi_g : x \mapsto \max_{i=1,\dots,m}\langle x, y_i\rangle + g_i - \frac{\|y_i\|^2}{2}$ for which $\hat{T}_{n,m}$ is a selection of its subdifferential. This function is globally R-Lipschitz (in particular $2R$-Lipschitz on B_{2R}). We can thus apply Lemma 3.6 to this Brenier potential along with Jensen's inequality

and obtain

$$\left[\mathbb{E}\left(\int_{\mathbb{R}^d} \|\nabla\bar{\varphi}(x) - \hat{T}_{n,m}(x)\| d\mu(x)\right)\right]^4 \leq \mathbb{E}\left[\left(\int_{\mathbb{R}^d} \|\nabla\bar{\varphi}(x) - \hat{T}_{n,m}(x)\| d\mu(x)\right)^4\right]$$
$$\leq C^4 \mathbb{E}\left[(W_1(\mu, \hat{\mu}_n) + W_1(\nu, \hat{\nu}_m))\right]$$

where C is the multiplicative constant of Corollary 4.8. We now apply [Fournier and Guillin, 2015, Main Theorem]. For any integer $q \geq 3$ we have the following inequality

$$\mathbb{E}\left[W_1(\mu, \hat{\mu}_n)\right] \leq K_{d,q}\left(\int_{\mathbb{R}^d} \|x\|^q d\mu(x)\right)^{1/q} \times \begin{cases} n^{-1/2} + n^{-1+1/q} & \text{if } d = 1, \\ n^{-1/2}\log(1+n) + n^{-1+1/q} & \text{if } d = 2, \\ n^{-1/d} + n^{-1+1/q} & \text{if } d > 2 \end{cases}$$

where $K_{d,q} > 0$ is a constant depending only on d and q given in [Fournier and Guillin, 2015]. We choose $q = 3$ for example and upper bound the dominated term $n^{-2/3}$ by $n^{-1/2}$. This yields the result.

□

Remark 3.8. *We also describe an L^2 error bound in Lemma 4.9. Therefore similar statistical rates as in Lemma 3.7 also hold in L^2, with an exponent $1/5$.*

Remark 3.9. *The previous theorem can easily be adapted to the case where only one measure is empirical. For example, in Section 3.4, we consider the semi-discrete case in which μ is approximated by samples and ν is a fixed discrete probability measure.*

Remark 3.10. *Let us consider the case where μ admits an L^1 Poincaré inequality, in the sense of [Bakry et al., 2008, Equation (1.2)]. This holds true for example if μ has a log-concave density $\exp(-V)$ for a C^2, convex lower bounded V [Bakry et al., 2008, Corollary 1.9] or a bounded density away from 0 on a convex domain [Acosta and Durán, 2004]. In these cases, the explicit empirical Brenier potential in Lemma 3.1 converges to a Lipschitz optimal Brenier potential φ from Lemma 2.1, up to an additive term, at the same rate as in Lemma 3.7. Indeed, any Lipschitz function can be approximated, together with its gradient, by a smooth function. Lemma 3.7 allows to control the L^1 distance of gradients, and Poincaré inequality gives, up to an additive constant, a control of the L^1 distance of potentials. The resulting rates on convergence of potentials are probably suboptimal.*

3.3 Comparison with existing art on Monge map estimation

For simplicity, we discuss the case $n = m$ and $d > 2$ corresponding to case 3 of Lemma 3.7. Our convergence rate is related to that of the empirical Wasserstein distance. It has been established in [Fournier and Guillin, 2015] that $\mathbb{E}[W_1(\mu, \hat{\mu}_n)] = O\left(n^{-\frac{1}{d}}\right)$, and this rate is attained for the uniform distribution on the hypercube. Our rate is similar, up to the $1/4$ exponent, which arises from the stability result of Lemma 4.8 (see also Section 4.4). The literature on convergence rates of transport map estimators is vast, we start with [Manole et al., 2024] which describes $n^{-1/d}$ rates in L^2 norm. They are obtained under the following assumption.

Assumption 2 (Monge diffeomorphism assumption)**.** *Let μ and ν be two compactly supported measures. There exists an optimal Brenier potential in* (DP) *whose gradient is a bi-Lipschitz diffeomorphism.*

Remark 3.11. *In [Manole et al., 2024] this assumption is written as a condition on the Hessian of the potential given by* $\frac{1}{\lambda} I_d \preceq \nabla^2 \varphi \preceq \lambda I_d$*, for some* $\lambda > 0$*.*

The obtained rate in [Manole et al., 2024] is min-max optimal against the Monge diffeomorphisms class. Several estimators are proposed. One is based on one-nearest neighbor combined with barycentric projection, computationally efficient, but not the gradient of a convex function. The second is a least-squares estimator constrained to the class of gradients of convex functions, with a computational overhead compared to mere computation of the empirical Wasserstein distance, and in addition, for which λ needs to be known.

In this context, our estimator $\hat{T}_{n,m}$ and its analysis in Lemma 3.7 has substantial advantages described in the following.

1. Since $\hat{T}_{n,m}$ is the sub-gradient of a convex function, it serves as a natural estimator for Monge maps as it is the smallest class of functions containing all such maps.

2. It does not require any computation beyond dual optimal $g \in \mathbb{R}^m$ in Lemma 3.1 from the discrete dual problem. This is given as a byproduct for many solvers.

3. Our assumptions on the population Brenier potentials are much more applicable compared to Assumption 2. In particular we do not even require continuity of the Monge map. We are not aware of any quantitative analysis in this setting.

Yet, our analysis retains statistical rates similar to the Wasserstein rate up to the $1/4$ exponent which is independent of the dimension. It would be interesting to know if this rate could be further refined to $n^{-1/d}$ for our estimator under Assumption 2, this question is left for future work.

Beyond [Manole et al., 2024], let us mention other works. The first contribution to consider statistical estimation of Monge maps based on the dual transport problem is [Hütter and Rigollet, 2021], which also works under Assumption 2. The authors establish min-max results for transport map estimators whose convergence rate approaches $n^{-1/2}$ in L^2 norm for Monge maps of high differentiability order. These estimators become computationally expensive as the dimension grows. Another example is [Pooladian and Niles-Weed, 2021] which considers entropic transport maps, and [Divol et al., 2025] for estimation in abstract function spaces (e.g. shallow neural networks and RKHS), both under Assumption 2. We are not aware of any quantitative analysis beyond this assumption, in particular allowing for possibly discontinuous Monge transport maps, except [Pooladian et al., 2023] which is limited to the semi-discrete case.

Let us now discuss explicit sufficient conditions on the source and target measures ensuring that Assumption 2, given in abstract form, holds true. This is actually one of the important challenges in optimal transport beyond statistical estimation, see [Hütter and Rigollet, 2021, Remark 3], [Manole et al., 2024, Section 2] and [Villani, 2008, Section 12]. The most widely known sufficient conditions on the measures is Caffarelli's celebrated result on smoothness of Monge maps [Caffarelli, 1992a, Caffarelli, 1992b, Caffarelli, 1996] (summarized in [Villani, 2008, Theorem 12.50]). It states that, if both the source and target measures are supported on compact convex sets with Hölder continuous densities, bounded away from $+\infty$ and 0, then the Monge diffeomorphism assumption holds true. The scope of applicability is much narrower compared to ours: compactly supported measures, one of them with an upper bounded density.

3.4 The semi-discrete case

In this section, we consider an absolutely continuous source measure μ and a fixed discrete target measure ν, a setting known as semi-discrete optimal transport, as summarized in the following assumption.

Assumption 3. *Let $R > 0$ and $M > 0$, μ, ν be supported in B_R, such that μ has density upper-bounded by M. Assume further that $\nu = \sum_{j=1}^m b_j \delta_{y_i}$ for some fixed integer $m > 1$, support points $y_1, \dots, y_m \in \mathbb{R}^d$ and weight vector $b_1, \dots, b_m > 0$ such that $\sum_{j=1}^m b_j = 1$. For any positive integers n, consider i.i.d random samples $X_1, \dots, X_n$ distributed according to μ. We then define the empirical measure $\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}$. Our Monge map estimator is $\hat{T}_n$ as in Lemma 3.3 for discrete measures $\hat{\mu}_n$ and ν.*

Using similar ideas as in Lemma 3.2, the dual formulation in Lemma 2.1 in the semi-discrete setting has a finite dimensional equivalent expression. This type of argument is typical of the semi-discrete setting [Lévy, 2024] and is most often presented with Kantorovich potentials. We use here the dual framework of Lemma 2.1 and present proof details for completeness.

Lemma 3.12. *Let μ be a probability measure on $\mathbb{R}^d$, and ν be as in Assumption 3, both with compact support in B_R. For any $g \in \mathbb{R}^m$, let $\varphi_g \colon x \mapsto \max_{k=1,\dots,m} \langle x, y_k\rangle + g_k - \frac{1}{2}\|y_k\|^2$ as in Lemma 3.1. The following three minimization problems have the same value, each of them being attained:*

$$\min_{\varphi \in \Gamma_0(\mathbb{R}^d)} \int \varphi d\mu + \int \varphi^* d\nu \tag{3.2}$$

$$\min_{g \in \mathbb{R}^m} \int \varphi_g d\mu + \int \varphi_g^* d\nu \tag{3.3}$$

$$\min_{g \in \mathbb{R}^m} \int \varphi_g d\mu + \sum_{j=1}^m b_j \left(\frac{1}{2}\|y_j\|^2 - g_j\right). \tag{3.4}$$

Furthermore, for $g \in \mathbb{R}^m$ optimal in (3.4)*, we have $\varphi_g^*(y_j) = \frac{1}{2}\|y_j\|^2 - g_j$ for all $j = 1, \dots, m$, and g_j can be chosen in $[-3R^2, 3R^2]$, for all $j = 1, \dots, m$, with $g_m = 0$.*

Proof. For any $g \in \mathbb{R}^m$ and $\varphi_g \in \Gamma_0(\mathbb{R}^d)$, we have for all $j = 1, \dots, m$

$$\begin{aligned}
\varphi_g^*(y_j) &= \sup_{x \in \mathbb{R}^d} \langle y_j, x\rangle - \varphi_g(x) \\
&= \sup_{x \in \mathbb{R}^d} \langle y_j, x\rangle - \max_{k=1,\dots,m} \left\{\langle x, y_k\rangle + g_k - \frac{1}{2}\|y_k\|^2\right\} \\
&= \frac{1}{2}\|y_j\|^2 - g_j + \sup_{x \in \mathbb{R}^d} \langle y_j, x\rangle + g_j - \frac{1}{2}\|y_j\|^2 - \max_{k=1,\dots,m} \left\{\langle x, y_k\rangle + g_k - \frac{1}{2}\|y_k\|^2\right\} \\
&\leq \frac{1}{2}\|y_j\|^2 - g_j. \qquad (3.5)
\end{aligned}$$

Furthermore, if there is x such that $\varphi_g(x) = \langle x, y_j\rangle + g_j - \frac{1}{2}\|y_j\|^2$, then (3.5) is actually an equality.

Let us show that (3.2) = (3.3). Lemma 2.1 ensures that the infimum in (3.2) is attained. Furthermore, we obviously have (3.3) $\geq$ (3.2). Consider an optimal $\varphi \in \Gamma_0(\mathbb{R}^d)$ for (3.2). It must hold that $\max_j \varphi^*(y_j) < +\infty$. For all j, set $g_j = \frac{1}{2}\|y_j\|^2 - \varphi^*(y_j)$ we have

$$\begin{aligned}
\varphi(x) = \sup_y \langle x, y\rangle - \varphi^*(y) \geq \max_{k=1,\dots,m} \langle x, y_k\rangle - \varphi^*(y_k) = \varphi_g(x) \qquad &\forall x \in \mathbb{R}^d, \\
\varphi^*(y_j) = \frac{1}{2}\|y_j\|^2 - g_j \geq \varphi_g^*(y_j) \qquad &\forall j = 1, \dots, m,
\end{aligned}$$

where we used (3.5). We deduce that (3.2) $\geq$ (3.3), so that both value are equal, and the infimum in (3.3) is also attained.

Let us now show that (3.3) = (3.4). First, (3.5) ensures that (3.3) $\leq$ (3.4). The proof of equality follows by the following claim, for example applied to an optimal $\bar{g}$ in (3.3).

Claim 3.13. *For any $\bar{g} \in \mathbb{R}^m$, there exists $g \geq \bar{g}$ coordinatewise such that $\varphi_g = \varphi_{\bar{g}}$ and* (3.5) *is an equality for φ_g for all $j = 1, \ldots, m$. Furthermore, $g_j > \bar{g}_j$ for all j such that* (3.5) *is not an equality for $\varphi_{\bar{g}}$.*

Let us now prove the claim. Set $\bar{J} = \{j = 1, \ldots, m | \, \exists x \in \mathbb{R}^d, \varphi_{\bar{g}}(x) = \langle x, y_j \rangle + \bar{g}_j - \frac{1}{2}\|y_j\|^2\}$. For all $j \in \bar{J}$, (3.5) is an equality and we may set x_j such that $\varphi_{\bar{g}}(x_j) = \langle x_j, y_j \rangle + \bar{g}_j - \frac{1}{2}\|y_j\|^2$. Fix any $q \notin \bar{J}$, it holds that (3.5) is strict, and actually

$$\min_x \varphi_{\bar{g}}(x) - \langle x, y_q \rangle - \bar{g}_q + \frac{1}{2}\|y_q\|^2 > 0,$$

since the infimum of a polyhedral convex function is attained (see [Rockafellar, 1970, Chapter 19], the infimum is expressed as a bounded linear program whose value is always attained). We may denote the argmin by $\bar{x}$ and set

$$g_q = \min_x \varphi_{\bar{g}}(x) - \langle x, y_q \rangle + \frac{1}{2}\|y_q\|^2 = \varphi_{\bar{g}}(\bar{x}) - \langle \bar{x}, y_q \rangle + \frac{1}{2}\|y_q\|^2 > \bar{g}_q,$$

and $g_k := \bar{g}_k$ for $k \neq q$. We have $\varphi_{\bar{g}} \leq \varphi_g$ by construction, since $g \geq \bar{g}$ coordinatewise.

For any x such that $\varphi_g(x) > \langle x, y_q \rangle + g_q - \frac{1}{2}\|y_q\|^2$, we have

$$\varphi_g(x) = \max_{k \neq q} \langle x, y_k \rangle + g_k - \frac{1}{2}\|y_k\|^2 = \max_{k \neq q} \langle x, y_k \rangle + \bar{g}_k - \frac{1}{2}\|y_k\|^2 \leq \max_k \langle x, y_k \rangle + \bar{g}_k - \frac{1}{2}\|y_k\|^2 = \varphi_{\bar{g}}(x).$$

For any x such that $\varphi_g(x) = \langle x, y_q \rangle + g_q - \frac{1}{2}\|y_q\|^2$, we have

$$\varphi_g(x) = \langle x, y_q \rangle + \min_z \varphi_{\bar{g}}(z) - \langle z, y_q \rangle + \frac{1}{2}\|y_q\|^2 - \frac{1}{2}\|y_q\|^2 \leq \varphi_{\bar{g}}(x).$$

Thus $\varphi_g = \varphi_{\bar{g}}$.

Finally, we have

$$\begin{aligned}
\varphi_g(\bar{x}) &= \max_{k=1,\ldots,m} \langle \bar{x}, y_k \rangle + g_k - \frac{1}{2}\|y_k\|^2 \geq \langle \bar{x}, y_q \rangle + g_q - \frac{1}{2}\|y_q\|^2 \\
&= \langle \bar{x}, y_q \rangle + \varphi_{\bar{g}}(\bar{x}) - \langle \bar{x}, y_q \rangle + \frac{1}{2}\|y_q\|^2 - \frac{1}{2}\|y_q\|^2 = \varphi_{\bar{g}}(\bar{x}) = \varphi_g(\bar{x}),
\end{aligned}$$

and thus

$$\varphi_g(\bar{x}) = \langle \bar{x}, y_q \rangle + g_q - \frac{1}{2}\|y_q\|^2. \tag{3.6}$$

In addition for all $j \in \bar{J}$

$$\varphi_g(x_j) = \varphi_{\bar{g}}(x_j) = \langle x_j, y_j \rangle + \bar{g}_j - \frac{1}{2}\|y_j\|^2 = \langle x_j, y_j \rangle + g_j - \frac{1}{2}\|y_j\|^2. \tag{3.7}$$

We deduce by combining (3.6) and (3.7) that (3.5) holds with equality for φ_g, for all $j \in \bar{J} \cup \{q\}$. We may repeat this process recursively for each $q \notin \bar{J}$ and obtain $\varphi_g = \varphi_{\bar{g}}$ such that (3.5) is an equality for all j. This shows (3.3) = (3.4) and concludes the proof.

We now prove the last statement of the Lemma. Let $g \in \mathbb{R}^m$ be optimal in (3.4) such that (3.5) is an equality for all j. The addition of a constant to all the coordinates of g does not change the functional value in (3.4), we may therefore assume that $g_m = 0$. By Lemma 2.2 there exists a function in $\mathcal{C}_{R,R}$ equal to φ_g over

$\operatorname{supp}\mu$ with conjugate equal to φ_g^* over $\operatorname{supp}\nu$. For any $i \in \{1, \dots, m\}$, using that $\varphi_g^*(y_i) = \frac{1}{2}\|y_i\|^2 - g_i$ and the fact that $y_i, y_m \in \operatorname{supp}\nu$ we obtain

$$|g_i| = |g_i - g_m| = \left|\frac{1}{2}\|y_i\|^2 - \frac{1}{2}\|y_m\|^2 - \varphi_g^*(y_i) + \varphi_g^*(y_m)\right| \leq R^2 + R\|y_i - y_m\| \leq 3R^2.$$

□

The finite-dimensional parametrization in Lemma 3.12 allows us to adapt Lemma 3.7, yielding an improved statistical rate, with dimension free-exponent.

Theorem 3.14. *For all $M > 0$, $R > 0$, and $d \in \mathbb{N}^*$ there exists $C > 0$ such that for all μ, ν as in Assumption 3 the following holds. Let $\bar{\varphi}$ be an optimal Brenier potential in* (DP) *for measures μ and ν. Then, for ν fixed,*

$$\mathbb{E}\left[\int_{\mathbb{R}^d} \|\nabla\bar{\varphi}(x) - \hat{T}_n(x)\| d\mu(x)\right] \leq C\left(\frac{m}{n}\right)^{\frac{1}{8}}.$$

Proof. Invoking Lemma 3.12, we have $\bar{g}, \hat{g}$ optimal for (3.4), with infinite norm at most $3R^2$, associated respectively to the semi-discrete problem for (μ, ν) and the discrete problem for $(\hat{\mu}_n, \nu)$. The resulting potentials, $\varphi_{\bar{g}}$ and $\varphi_{\hat{g}}$ are optimal for (3.3) and are therefore optimal dual potentials in Lemma 2.1. Finally we may consider that (3.5) holds as equality for all $i = 1, \dots, m$ for both $\varphi_{\bar{g}}$ and $\varphi_{\hat{g}}$. We have

$$\begin{aligned}
&\int_{\mathbb{R}^d} \varphi_{\hat{g}} d\mu + \int_{\mathbb{R}^d} \varphi_{\hat{g}}^* d\nu - \left(\int_{\mathbb{R}^d} \bar{\varphi} d\mu + \int_{\mathbb{R}^d} \bar{\varphi}^* d\nu\right) \\
&= \int_{\mathbb{R}^d} \varphi_{\hat{g}} d\mu + \sum_{i=1}^m b_i \left(\frac{1}{2}\|y_i\|^2 - \hat{g}_i\right) - \left(\int_{\mathbb{R}^d} \varphi_{\bar{g}} d\mu + \sum_{i=1}^m b_i \left(\frac{1}{2}\|y_i\|^2 - \bar{g}_i\right)\right) \\
&\leq \int_{\mathbb{R}^d} \varphi_{\hat{g}} d\mu + \sum_{i=1}^m b_i \left(\frac{1}{2}\|y_i\|^2 - \hat{g}_i\right) - \left(\int_{\mathbb{R}^d} \varphi_{\hat{g}} d\hat{\mu}_n + \sum_{i=1}^m b_i \left(\frac{1}{2}\|y_i\|^2 - \hat{g}_i\right)\right) \\
&\quad + \int_{\mathbb{R}^d} \varphi_{\bar{g}} d\hat{\mu}_n + \sum_{i=1}^m b_i \left(\frac{1}{2}\|y_i\|^2 - \bar{g}_i\right) - \left(\int_{\mathbb{R}^d} \varphi_{\bar{g}} d\mu + \sum_{i=1}^m b_i \left(\frac{1}{2}\|y_i\|^2 - \bar{g}_i\right)\right) \\
&\leq 2 \sup_{g \in [-3R^2, 3R^2]^m} \left|\int_{\mathbb{R}^d} \varphi_g d\mu - \int_{\mathbb{R}^d} \varphi_g d\hat{\mu}_n\right|
\end{aligned}$$

where we used for the first inequality the fact that $\hat{g}$ is a minimizer in (3.4) for measure $\hat{\mu}_n$. We apply Lemma 4.7 to the potentials $\bar{\varphi}$ and $\varphi_{\hat{g}}$ and Jensen's inequality. Recalling that $\hat{T}_n = \nabla\varphi_{\hat{g}}$ almost everywhere, it yields

$$\frac{1}{C^4}\mathbb{E}\left[\int \|\nabla\bar{\varphi}(x) - \hat{T}_n(x)\| d\mu(x)\right]^4 \leq \mathbb{E}\left[2 \sup_{g \in [-3R^2, 3R^2]^m} \left|\int_{\mathbb{R}^d} \varphi_g d\mu - \int_{\mathbb{R}^d} \varphi_g d\hat{\mu}_n\right|\right]. \tag{3.8}$$

To upper bound the expectation of the last term we are going to use [Giné and Nickl, 2021, Theorem 3.5.1] with the following class of functions

$$\mathcal{F} = \{x \mapsto \varphi_g(x),\ g \in \left[-3R^2, 3R^2\right]^m\}.$$

Notice that for all $x \in \mathbb{R}^d$ and for any $w, v \in \mathbb{R}^m$ we have $|\varphi_w(x) - \varphi_v(x)| \leq \|w - v\|_\infty$ and thus $\|\varphi_w - \varphi_v\|_{L^2(\mu_n)} \leq \|w - v\|_\infty$.

We thus verify the following chain of inequalities for covering numbers

$$\mathcal{N}(\tau/2, \mathcal{F}, \|\cdot\|_{L^2(\mu_n)}) \leq \mathcal{N}(\tau/2, [-3R^2, 3R^2]^m, \|\cdot\|_\infty) \leq \mathcal{N}(\tau/2, \left[-3R^2, 3R^2\right], |\cdot|)^m \leq \left(\frac{12R^2}{\tau}\right)^m.$$

Furthermore, we have for all $x \in B_R$ and for all $g \in [-3R^2, 3R^2]^m$ that $|\varphi_g(x)| \leq \frac{9}{2}R^2$ and thus $\sup_{g \in [-3R^2, 3R^2]^m} \|\varphi_g\|_{L^2(\mu)} \leq \frac{9}{2}R^2$. Combining (3.8) with [Giné and Nickl, 2021, Theorem 3.5.1] we then obtain the desired result

$$\frac{1}{C^4}\mathbb{E}\left[\int \|\nabla\bar{\varphi}(x) - \hat{T}_n(x)\| d\mu(x)\right]^4 \leq 16\sqrt{2}\sqrt{\frac{m}{n}}\int_0^{\frac{9}{2}R^2} \sqrt{\log\left(\frac{24R^2}{\tau}\right)} d\tau.$$

□

Remark 3.15. *The previous theorem can be extended to the case where ν is also approximated by samples, yielding the same convergence rate. The proof can be derived from a conditioning argument, as in [Pooladian et al., 2023, Appendix F].*

Remark 3.16. *Lemma 3.10, which discusses the case where μ satisfies an L^1 Poincaré inequality, also applies to the previous theorem.*

Comparison with existing semi-discrete Monge map estimators A semi-discrete Monge map estimator based on the entropic optimal transport map between the empirical source and target measures is studied in [Pooladian et al., 2023]. Similarly to Section 3.2, their analysis assumes that the source population measure admits a density that is bounded both from above and below on its convex compact support. Under this assumption, they establish convergence rate in $L_2(\mu)$ of order $O\left((m/n)^{\frac{1}{4}}\right)$ which they show to be minimax optimal in n.

By contrast, our analysis results in a convergence rate of order $O\left((m/n)^{\frac{1}{8}}\right)$ in $L^1(\mu)$ without requiring convexity of the support nor that the density of the source measure is lower bounded away from zero. We do not know whether this rate is minimax optimal under these general assumptions. Moreover, it remains unclear whether our convergence rate analysis could be improved under the stronger regularity assumptions considered in [Pooladian et al., 2023]. Addressing both questions is left for future work. Finally, our estimator compares favorably to the baseline, one nearest neighbor estimator, which suffers from the curse of dimensionality in the semi-discrete setting, as described in [Pooladian et al., 2023].

3.5 Convergence of couplings

Existing works on the convergence of empirical optimal couplings are asymptotic [Seguy et al., 2018, Segers, 2022], and we are not aware of finite sample bounds. The following result provides such a bound, based on a stability estimate for optimal couplings presented in Section 5.

Theorem 3.17. *For any $M > 0,\ R > 0$ and $d \in \mathbb{N}^*$ under the setting of Assumption 1, there exists $C > 0$ such that the following holds. Let $\hat{\gamma}_{n,m}$ be an optimal coupling for measures $\hat{\mu}_n, \hat{\nu}_m$, and γ be an optimal*

coupling for measures μ, ν, then

$$\mathbb{E}\left[W_2(\gamma, \hat{\gamma}_{n,m})\right] \leq C \begin{cases} \left(n^{-1/4} + m^{-1/4}\right)^{1/8} & \text{if } d < 4, \\ \left(n^{-1/4}\sqrt{\log(1+n)} + m^{-1/4}\sqrt{\log(1+m)}\right)^{1/8} & \text{if } d = 4, \\ \left(n^{-1/d} + m^{-1/d}\right)^{1/8} & \text{if } d > 4. \end{cases}$$

Proof. We apply Lemma 5.4 to coupling γ and $\hat{\gamma}_{n,m}$. There exists $C > 0$ depending on d, R and M such that $W_2(\gamma, \hat{\gamma}_{n,m})^8 \leq C(W_2(\mu, \hat{\mu}_n) + W_2(\nu, \hat{\nu}_m))$. As in Lemma 3.7, we apply [Fournier and Guillin, 2015, Main Theorem] for W_2, with $q = 4$, yielding the result. Note that an additional $\frac{1}{2}$ factor arises as a byproduct of considering $W_2(\mu, \hat{\mu}_n)$ (resp. $W_2(\nu, \hat{\nu}_m)$) instead of $W_2^2(\mu, \hat{\mu}_n)$ (resp. $W_2^2(\nu, \hat{\nu}_m)$) in [Fournier and Guillin, 2015]. □

Remark 3.18. *This estimate is unlikely to admit significant improvement in the semi-discrete setting. Indeed, as suggested by the stability result for couplings on which Lemma 3.17 builds, the convergence of the couplings cannot be substantially faster than that of the marginals. Since $W_2(\mu, \hat{\mu}_n)$ is subject to the curse of dimensionality, this is also the case for $W_2(\gamma, \hat{\gamma}_{n,m})$.*

4 An error bound for Brenier dual potentials

4.1 Preliminary results

We recall the following [Carlier et al., 2025, Theorem 2.1] on the covering number of singularities of convex functions.

Theorem 4.1. *Let $\phi\colon \mathbb{R}^d \to \mathbb{R}$ be convex and $L, R > 0$, such that ϕ is L-Lipschitz on $B(0, R)$. Set $\Sigma_{\eta,\alpha} = \left\{x \in \mathbb{R}^d | \operatorname{diam}(\partial\phi(B(x, \eta))) \geq \alpha\right\}$. Then for all α and $\eta > 0$, we have the following bound on the covering number*

$$\mathcal{N}(\Sigma_{\eta,\alpha} \cap B(0, R), 8\eta) \leq 48d^2(R + 4\eta)^{d-1}\frac{L}{\alpha\eta^{d-1}}.$$

Note that this theorem slightly differs from the formulation in [Carlier et al., 2025, Theorem 2.1] which requires a global Lipschitz modulus for ϕ. Still, it is equivalent since the covering number is restricted to $B(0, R)$. Using [Cobzas and Mustata, 1978], the function ϕ may be extended to an L Lipschitz convex function on the whole space, fitting the formulation in [Carlier et al., 2025]. We also recall the definition of the proximal operator on $\mathbb{R}^d$ [Moreau, 1965].

Definition 4.2. *For any $f \in \Gamma_0(\mathbb{R}^d)$, the proximal operator is defined, for all $v \in \mathbb{R}^d$, as*

$$\operatorname{prox}_f(v) = \operatorname*{argmin}_{x \in \mathbb{R}^d} f(x) + \frac{1}{2}\|x - v\|^2.$$

Finally, we recall the Fenchel-Young inequality with a remainder due to [Carlier, 2023].

Lemma 4.3. *Let $\phi \in \Gamma_0(\mathbb{R}^d)$. For every $x, p \in \mathbb{R}^d$ and $\lambda > 0$ we have*

$$\phi(x) + \phi^*(p) - \langle x, p\rangle \geq \frac{1}{\lambda}\|x - \mathrm{prox}_{\lambda\phi}(x + \lambda p)\|^2.$$

Extensions around Lemma 4.3 can be found in [Bauschke et al., 2022], [Burachik and Martínez-Legaz, 2025] and in [Combettes and Mayrand, 2026] with a historical discussion.

4.2 A dual error bound

Leveraging the results presented in Section 4.1, we obtain the following error bound for convex potentials, relating an integrated Fenchel-Young remainder to L_1 distance of gradients. In an optimal transport context, this provides an error bound on the dual problem.

Theorem 4.4. *Let μ be absolutely continuous with density upper bounded by M on $\mathbb{R}^d$, compactly supported in B_R and $\bar{\varphi}, \phi$ be convex lower semi-continuous on $\mathbb{R}^d$ and L-Lipschitz on B_{2R}. Consider any $q \in [1, 2]$ and the following constant:*

$$C_{R,d} = 4\frac{e}{\sqrt{2\pi}\sqrt{\frac{d}{2}+1}}\left(\frac{\pi e 64}{(\frac{d}{2}+1)}\right)^{\frac{d}{2}} 48d^2(2R)^{d-1}.$$

Case 1: *If*

$$\int \phi(x) + \phi^*(\nabla\bar{\varphi}(x)) - \langle x, \nabla\bar{\varphi}(x)\rangle \, d\mu(x) > \left(\frac{R}{8}\right)^{\frac{q+3}{q+1}} 2L(MC_{R,d})^{\frac{2}{q+1}}$$

then

$$\begin{aligned}\|\nabla\bar{\varphi} - \nabla\phi\|_{L^q(\mu)} \leq & 4\sqrt{\frac{L}{R}}\left(\int \phi(x) + \phi^*(\nabla\bar{\varphi}(x)) - \langle x, \nabla\bar{\varphi}(x)\rangle \, d\mu(x)\right)^{\frac{1}{2}} \\ & + L\left(\frac{32}{RL}\int \phi(x) + \phi^*(\nabla\bar{\varphi}(x)) - \langle x, \nabla\bar{\varphi}(x)\rangle \, d\mu(x)\right)^{\frac{q+1}{2q}}.\end{aligned}$$

Case 2: *Otherwise*

$$\|\nabla\bar{\varphi} - \nabla\phi\|_{L^q(\mu)} \leq 4\left(L^{q+2}MC_{R,d}\right)^{\frac{1}{q+3}}\left(\int \phi(x) + \phi^*(\nabla\bar{\varphi}(x)) - \langle x, \nabla\bar{\varphi}(x)\rangle \, d\mu(x)\right)^{\frac{1}{q+3}}.$$

Remark 4.5. *The following equivalent expressions explicitly establish the connection with the dual problem of optimal transport in* (1.2)*, where ν is the pushforward of μ by $\nabla\bar{\varphi}$. This is a consequence of the fact that $\langle x, \nabla\bar{\varphi}(x)\rangle = \bar{\varphi}(x) + \bar{\varphi}^*(\nabla\bar{\varphi}(x))$ for μ almost all x, and*

$$\begin{aligned}&\int \phi(x) + \phi^*(\nabla\bar{\varphi}(x)) - \langle x, \nabla\bar{\varphi}(x)\rangle \, d\mu(x) \\ &= \int \phi(x) + \phi^*(\nabla\bar{\varphi}(x))d\mu(x) - \int \bar{\varphi}(x) + \bar{\varphi}^*(\nabla\bar{\varphi}(x))d\mu(x) = \int \phi d\mu + \int \phi^* d\nu - \left(\int \bar{\varphi}d\mu + \int \bar{\varphi}^* d\nu\right).\end{aligned}$$

In this case, the right hand side is the dual suboptimality gap of the optimal transport problem between μ and ν in Lemma 2.1.

Proof. For any $\lambda > 0$ (to be chosen later), and μ almost all x, set $z_\lambda(x) = \mathrm{prox}_{\lambda\phi}(x + \lambda\nabla\bar{\varphi}(x))$. This is defined by the relation $z_\lambda(x) = (I + \lambda\partial\phi)^{-1}(x + \lambda\nabla\bar{\varphi}(x))$ or equivalently,

$$v_\lambda(x) := \nabla\bar{\varphi}(x) + \frac{x - z_\lambda(x)}{\lambda} \in \partial\phi(z_\lambda(x)). \tag{4.1}$$

We have $\mathrm{prox}_{\lambda\phi}(x + \lambda v) = x$ for any $v \in \partial\phi(x)$. Moreover, the prox mapping being 1-Lipschitz, we deduce that for all $x \in B(0, R)$,

$$\|z_\lambda(x) - x\| = \|\mathrm{prox}_{\lambda\phi}(x + \lambda\nabla\bar{\varphi}(x)) - \mathrm{prox}_{\lambda\phi}(x + \lambda v)\| \leq \lambda\|\nabla\bar{\varphi}(x) - v\| \leq 2\lambda L := \eta, \tag{4.2}$$

where η is a constant appearing in the covering number in Lemma 4.1 to be chosen later. For the last inequality, we have used the fact that both $\bar{\varphi}$ and ϕ are L-Lipschitz in a neighborhood of any $x \in \mathrm{supp}\ \mu$. We are going to consider only values $\eta \leq R$, so that $z_\lambda(x) \in B(0, 2R)$ for any $x \in \mathrm{supp}\ \mu$, and the corresponding $v_\lambda(x)$ is bounded by L using the Lipschitz continuity of ϕ and the characterization (4.1). Set

$$\delta = \int \phi(x) + \phi^*(\nabla\bar{\varphi}(x)) - \langle x, \nabla\bar{\varphi}(x)\rangle\, d\mu(x).$$

By Carlier's inequality in Lemma 4.3, we have for any $\lambda > 0$, using (4.1),

$$\delta \geq \frac{1}{\lambda}\int \|x - z_\lambda(x)\|^2 d\mu(x) = \lambda \int \|\nabla\bar{\varphi}(x) - v_\lambda(x)\|^2 d\mu(x).$$

We deduce, using Jensen inequality with $q \leq 2$, and introducing $\mathbb{I}$ the function that maps a boolean value true/false to the corresponding numerical value 1/0,

$$\|\nabla\bar{\varphi} - v_\lambda\|^2_{L^q(\mu)} \leq \left(\int \|\nabla\bar{\varphi}(x) - v_\lambda(x)\|^2 d\mu(x)\right)^{\frac{q}{2}\times\frac{2}{q}} = \int \|\nabla\bar{\varphi}(x) - v_\lambda(x)\|^2 d\mu(x) \leq \frac{\delta}{\lambda}. \tag{4.3}$$

For any $\alpha \geq 0$ (appearing in the covering number in Lemma 4.1, to be chosen later), and $\eta = 2L\lambda \leq R$, we have using the definition of v_λ in (4.1),

$$\begin{aligned}
\int \|v_\lambda - \nabla\phi\|^q d\mu &\leq (2L)^q \int \mathbb{I}(\|v_\lambda(x) - \nabla\phi(x)\| \geq \alpha) d\mu(x) + \alpha^q \int \mathbb{I}(\|v_\lambda(x) - \nabla\phi(x)\| < \alpha) d\mu(x) \\
&\leq (2L)^q \int \mathbb{I}(\mathrm{diam}(\partial\phi(B(x, \eta))) \geq \alpha) d\mu(x) + \alpha^q \\
&\leq (2L)^q M\ \mathrm{Vol}(B(0, 8\eta))\, \mathcal{N}(\Sigma_{\eta,\alpha} \cap B(0, R), 8\eta) + \alpha^q,
\end{aligned} \tag{4.4}$$

where the first inequality is because both v_λ and $\nabla\phi$ are bounded by L on $\mathrm{supp}\ \mu$, the second is because of the characterization (4.1) and the upper bound (4.2), and the last one is a covering argument combined with the fact that μ has a density bounded by M, $\Sigma_{\eta,\alpha}$ is given in Lemma 4.1.

We have, combining (4.3) and (4.4), for any $\eta \in (0, R)$ and $\alpha \geq 0$, using $2L\lambda = \eta$

$$\begin{aligned}
\|\nabla\bar{\varphi} - \nabla\phi\|_{L^q(\mu)} &\leq \|\nabla\bar{\varphi} - v_\lambda\|_{L^q(\mu)} + \|v_\lambda - \nabla\phi\|_{L^q(\mu)} \\
&\leq \sqrt{\frac{2L\delta}{\eta}} + \left((2L)^q M\ \mathrm{Vol}(B(0, 8\eta))\, \mathcal{N}(\Sigma_{\eta,\alpha} \cap B(0, R), 8\eta) + \alpha^q\right)^{\frac{1}{q}} \\
&\leq \sqrt{\frac{2L\delta}{\eta}} + 2L\left(M\ \mathrm{Vol}(B(0, 8\eta))\, \mathcal{N}(\Sigma_{\eta,\alpha} \cap B(0, R), 8\eta)\right)^{\frac{1}{q}} + \alpha,
\end{aligned} \tag{4.5}$$

where we have used the subaditivity of $t \mapsto t^{1/q}$ for $q \geq 1$.

We are going to work the two cases separately. Let us consider the first case $\delta > \left(\frac{R}{8}\right)^{\frac{q+3}{q+1}} 2L(MC_{R,d})^{\frac{2}{q+1}} \geq \frac{R}{32} L(MC_{R,d}R)^{\frac{2}{q+1}}$. We choose $\eta = \frac{R}{8}$ and $\alpha = 0$. In this case the covering number in (4.5) is $\mathcal{N}(\Sigma_{\eta,\alpha} \cap B(0,R), 8\eta) = 1$. We expand the right hand side of (4.5), using the volume estimates of Lemma A.2

$$4\sqrt{\frac{L\delta}{R}} + L\left(2^q M \operatorname{Vol}(B(0,R))\right)^{\frac{1}{q}} \leq 4\sqrt{\frac{L\delta}{R}} + L\left(M \frac{2^q e}{\sqrt{2\pi}\sqrt{\frac{d}{2}+1}} \left(\frac{\pi e}{\left(\frac{d}{2}+1\right)}\right)^{\frac{d}{2}} R^d\right)^{\frac{1}{q}}$$
$$\leq 4\sqrt{\frac{L\delta}{R}} + L(MC_{R,d}R)^{\frac{1}{q}} \leq 4\sqrt{\frac{L\delta}{R}} + L\left(\frac{32\delta}{RL}\right)^{\frac{q+1}{2q}}.$$

This concludes the first point.

Let us now consider the second case: $\delta \leq \left(\frac{R}{8}\right)^{\frac{q+3}{q+1}} 2L(MC_{R,d})^{\frac{2}{q+1}}$. Using Lemma 4.1 and the volume estimate in Lemma A.2, (4.5) becomes, for any $\eta \in]0, R[$ and $\alpha \geq 0$,

$$\|\nabla\bar{\varphi} - \nabla\phi\|_{L^q(\mu)} \leq L\left(M \frac{2^q e}{\sqrt{2\pi}\sqrt{\frac{d}{2}+1}} \left(\frac{\pi e 64}{\left(\frac{d}{2}+1\right)}\right)^{\frac{d}{2}} 48d^2(R+4\eta)^{d-1}\frac{\eta L}{\alpha}\right)^{\frac{1}{q}} + \alpha + \sqrt{\frac{2L\delta}{\eta}}.$$

For any $\eta \leq \frac{R}{4}$, we get $R + 4\eta \leq 2R$ so that, for any $\alpha \geq 0$,

$$\|\nabla\bar{\varphi} - \nabla\phi\|_{L^q(\mu)} \leq L\left(LMC_{R,d}\frac{\eta}{\alpha}\right)^{\frac{1}{q}} + \alpha + \sqrt{\frac{2L\delta}{\eta}}$$

Consider $\alpha = L\left(MC_{R,d}\eta\right)^{\frac{1}{q+1}}$, we have

$$\|\nabla\bar{\varphi} - \nabla\phi\|_{L^q(\mu)} \leq L\left((MC_{R,d}\eta)^{1-\frac{1}{q+1}}\right)^{\frac{1}{q}} + L\left(MC_{R,d}\eta\right)^{\frac{1}{q+1}} + \sqrt{\frac{2L\delta}{\eta}} = 2L(MC_{R,d}\eta)^{\frac{1}{q+1}} + \sqrt{\frac{2L\delta}{\eta}}$$

Choosing $\eta = \left(\frac{\sqrt{\delta}}{\sqrt{2L}(MC_{R,d})^{\frac{1}{q+1}}}\right)^{\frac{2q+2}{q+3}} \leq \frac{R}{8}$, we compute (details in Lemma 4.6)

$$\|\nabla\bar{\varphi} - \nabla\phi\|_{L^q(\mu)} \leq 2(2L)^{\frac{q+2}{q+3}}(MC_{R,d}\delta)^{\frac{1}{q+3}} \leq 4\left(L^{q+2}MC_{R,d}\delta\right)^{\frac{1}{q+3}}.$$

This concludes the proof of the second point. □

Lemma 4.6. *Assume $\delta, L, M, C > 0$ and $q \in [1,2]$. For $\eta = \left(\frac{\sqrt{\delta}}{\sqrt{2L}(MC)^{\frac{1}{q+1}}}\right)^{\frac{2q+2}{q+3}}$, we have*

$$2L(MC\eta)^{\frac{1}{q+1}} + \sqrt{\frac{2L\delta}{\eta}} = 2(2L)^{\frac{q+2}{q+3}}(MC\delta)^{\frac{1}{q+3}}.$$

The proof of this Lemma can be found in Section A.6.

4.3 Simplification and application to stability of optimal transport maps

We now only consider the case $q = 1$. The following holds because under the appropriate assumptions, all quantities of interest are bounded so that the $\frac{1}{q+3} = \frac{1}{4}$ exponent dominates over the exponent $\frac{q+1}{2q} = 1$ in Lemma 4.4, see also Lemma 4.5.

Corollary 4.7. *For any d, R, M, there is $C > 0$ such that the following holds. Let μ, ν be probability measures compactly supported in B_R such that $\mu(A) \leq M\lambda(A)$ for any measurable set $A \subset \mathbb{R}^d$. Let $\bar{\varphi}$ be an R-Lipschitz Brenier potential for μ and ν, optimal for* (DP) *as in Lemma 2.1. Then for any $\phi \in \mathcal{C}_{2R,2R}$ (as in Lemma 2.2), we have*

$$\int \|\nabla\bar{\varphi}(x) - \nabla\phi(x)\| d\mu(x) \leq C \left(\int \phi d\mu + \phi^* d\nu - \int \bar{\varphi} d\mu + \bar{\varphi}^* d\nu \right)^{\frac{1}{4}},$$

where $C = \max\left(268\sqrt{R}, 4(8R^3 M C_{R,d})^{1/4}\right)$ with $C_{R,d}$ defined in Lemma 4.4.

Proof. We want to establish an upper bound on $\delta := \int \phi d\mu + \int \phi^* d\nu - \left(\int \bar{\varphi} d\mu + \int \bar{\varphi}^* d\nu\right)$ using Lemma 4.4, see Lemma 4.5. Let $v \in \partial\phi(0)$, we have $\|v\| \leq 2R$ since ϕ is $2R$-Lipschitz on B_{2R}. Since ϕ^* is also $2R$ Lipschitz on B_{2R}, we have for all $y \in \operatorname{supp}\nu \subset B_R$:

$$\phi^*(y) = \phi^*(y) - \phi^*(v) + \phi^*(v) \leq 2R\|y - v\| + \phi^*(v) \leq 6R^2 + \phi^*(v).$$

We also have $0 \in \partial\phi^*(v)$, so that $v \in \operatorname{argmin}\phi^*$, and by the Fenchel-Young identity we obtain $\phi^*(v) = -\phi(0)$ and thus for all $x \in \operatorname{supp}\mu \subset B_R$ and all $y \in \operatorname{supp}\nu$:

$$\phi(x) + \phi^*(y) \leq \phi(x) + 6R^2 + \phi^*(v) = 6R^2 + \phi(x) - \phi(0) \leq 6R^2 + 2R\|x\| \leq 8R^2.$$

By Lemma 2.2, we may assume that $\bar{\varphi} \in \mathcal{C}_{2R,2R}$. We may repeat the same argument for $\bar{\varphi}$ to obtain $\delta \leq 16R^2 := \bar{\delta}$. Upon applying Theorem 4.4 for $q = 1$, we have in case 1, with $L = 2R$:

$$\int_{\mathbb{R}^d} \|\nabla\phi(x) - \nabla\bar{\varphi}(x)\| d\mu(x) \leq 4\sqrt{\frac{L}{R}}\delta^{1/2} + \frac{32}{R}\delta \leq \left(8\sqrt{2}\bar{\delta}^{1/4} + \frac{32}{R}\bar{\delta}^{3/4}\right)\delta^{1/4} \leq 268\sqrt{R}\delta^{1/4}$$

And in case 2, we have $\int_{\mathbb{R}^d} \|\nabla\phi(x) - \nabla\bar{\varphi}(x)\| d\mu(x) \leq 4(8R^3 M C_{R,d})^{1/4}\delta^{1/4}$, which concludes. □

Lemma 4.7 allows to obtain a stability bound for gradients of Brenier potentials.

Corollary 4.8. *For any d, R, M there is C positive such that for any $\mu, \nu, \bar{\varphi}$ as in Lemma 4.7, for any $\hat{\mu}$, $\hat{\nu}$ supported on B_R with $\phi_0 \in \Gamma_0(\mathbb{R}^d)$ an associated Brenier potential which is $2R$-Lipschitz on B_{2R}, we have*

$$\int \|\nabla\bar{\varphi}(x) - \nabla\phi_0(x)\| d\mu(x) \leq C \left(W_1(\mu, \hat{\mu}) + W_1(\nu, \hat{\nu})\right)^{\frac{1}{4}}.$$

Proof. From Lemma A.1, repeating the argument made in the proof of Lemma 2.2, there is a Brenier potential $\phi \in \mathcal{C}_{2R,2R}$, such that $\phi = \phi_0$ on B_{2R} and $\phi^* = \phi_0^*$ on $\operatorname{supp}\hat{\nu}$. We deduce that $\nabla\phi = \nabla\phi_0$ on

$\operatorname{supp}\mu$. Let $C>0$ be as in Lemma 4.7, we have

$$\begin{aligned}
&\frac{1}{C^4}\left(\int \|\nabla\bar\varphi(x)-\nabla\phi_0(x)\|d\mu(x)\right)^4 = \frac{1}{C^4}\left(\int \|\nabla\bar\varphi(x)-\nabla\phi(x)\|d\mu(x)\right)^4\\
&\leq \int \phi d\mu+\phi^* d\nu - \int \bar\varphi d\mu - \bar\varphi^* d\nu\\
&\leq \int \phi d\mu+\phi^* d\nu - \int \bar\varphi d\mu - \bar\varphi^* d\nu + \int \bar\varphi d\hat\mu + \bar\varphi^* d\hat\nu - \int \phi d\hat\mu - \phi^* d\hat\nu\\
&= \int \phi-\bar\varphi d(\mu-\hat\mu) + \int \phi^*-\bar\varphi^* d(\nu-\hat\nu) \;\leq\; 3R(W_1(\mu,\hat\mu)+W_1(\nu,\hat\nu)),
\end{aligned}$$

where the first inequality is by Lemma 4.7, the second is because ϕ is optimal for the dual problem associated to $\hat\mu$ and $\hat\nu$, and the last one is because $\phi-\bar\varphi$ and $\phi^*-\bar\varphi^*$ are both $3R$ Lipschitz so that [Villani, 2021, Theorem 1.14] can be applied. □

4.4 Connection with existing literature on stability of optimal transport map

Lemma 4.8 relates to the stability of gradient of Brenier potentials. This is most often stated in terms of $L^2(\mu)$ providing an Hilbertian embedding of the Wasserstein space. We present the L^2 version of Lemma 4.8 based on Lemma 4.4.

Corollary 4.9. *In the same setting as Lemma 4.8, there exists a constant $C>0$ such that*

$$\|\nabla\bar\varphi-\nabla\phi_0\|_{L^2(\mu)} \leq C\left(W_1(\mu,\hat\mu)+W_1(\nu,\hat\nu)\right)^{\frac{1}{5}}.$$

Sketch of proof. The proof follows the same reasoning as Lemma 4.7 and Lemma 4.8. The term $\delta := \int\phi d\mu+\int\phi^* d\nu-\left(\int\bar\varphi d\mu+\int\bar\varphi^* d\nu\right)$ can be bounded by $16R^2$ similarly to Lemma 4.8. Lemma 4.4 with $q=2$ leads to δ terms with exponents between $1/5$ and 1. Boundedness of δ and the same arguments as in Lemma 4.8 show that, up to constants the smallest exponent, $1/5$ dominates. □

This result generalizes the ideas of Monge embedding, extensively studied for a fixed source measure [Gigli, 2011, Mérigot et al., 2020, Delalande and Mérigot, 2023, Carlier et al., 2025]. Indeed Lemma 4.4 implies that

$$\|T_{\nu_1}-T_{\nu_2}\|_{L^2(\mu)} \leq CW_1(\nu_1,\nu_2)^{1/5} \tag{4.6}$$

where T_{ν_1}, T_{ν_2} are Monge maps between μ and ν_1, and μ and ν_2 respectively. In this Monge embedding context, Lemma 4.9 has several interesting properties: it allows both the source and target to move, the Hölder exponent is $1/5$ and it does not require convexity of the support or lower boundedness of the density (it does not require a Poincarré-Wirtinger inequality).

These can be discussed in light of existing literature. First, we are not aware of such stability results allowing both the source and target to move. Second [Delalande and Mérigot, 2023, Theorem 4.2] provides a $1/6$ exponent for μ supported on a convex compact set with density bounded away from zero and infinity. These conditions were relaxed in [Letrouit and Mérigot, 2024]. Let us emphasize that the cited results are deeper than the Monge embedding discussed in this section: they also provide stability for the Brenier potentials (not only their gradients), which requires a form of connectedness of the domain with nonvanishing density (hence the Poincarré inequality requirement).

Additionally, the exponent in Equation (4.6) cannot exceed $1/2$. This was shown in [Gigli, 2011], a different example being provided in [Mérigot et al., 2020, Proposition 2.1]. The $1/2$ exponent is achieved under Assumption 2, see [Hütter and Rigollet, 2021, Proposition 10]. Under broader assumptions, the exponent cannot exceed $1/4$ [Letrouit, 2025] which turns out to be optimal as shown by [Mérigot, 2026][1]. Finally, let us note that the constant and exponent in Lemma 4.8 or Lemma 4.9 directly appear in our main result Lemma 3.7.

5 A stability bound for optimal couplings

This section presents a stability result for couplings. We start with two technical lemmas and then present the main result of the section.

5.1 Preliminary results on perturbation and stability of optimal couplings

The following lemma allows to perturb the support of optimal couplings with a control in Wasserstein distance.

Lemma 5.1. *Let $\tilde{\mu}$ be compactly supported in B_R and $\tilde{\phi}, \bar{\varphi}$ be L Lipschitz on B_R. Assume that $\tilde{\nu}$ is another measure, such that $\tilde{\gamma}$ is the optimal coupling between $\tilde{\mu}$ and $\tilde{\nu}$ supported on the graph of $\partial\tilde{\phi}$. Then there is a coupling γ_0, supported on the graph of $\partial\bar{\varphi}$ with first marginal μ_0 supported in B_{2R}, such that*

$$W_2^2(\tilde{\gamma}, \gamma_0) \leq \frac{R^2 + 4L^2}{2LR}\left(\int \bar{\varphi} - \tilde{\phi} d\tilde{\mu} + \int \bar{\varphi}^* - \tilde{\phi}^* d\tilde{\nu}\right)$$
$$W_2^2(\tilde{\mu}, \mu_0) \leq \frac{R}{2L}\left(\int \bar{\varphi} - \tilde{\phi} d\tilde{\mu} + \int \bar{\varphi}^* - \tilde{\phi}^* d\tilde{\nu}\right)$$

Proof. Fix any $x, y \in \operatorname{supp}\ \tilde{\gamma}$, $\lambda > 0$, set $z_\lambda(x,y) = \operatorname{prox}_{\lambda\bar{\varphi}}(x + \lambda y)$ and $T(x,y) = (z_\lambda(x,y), y + (x - z_\lambda(x,y))/\lambda)$, characterized as

$$y + \frac{x - z_\lambda(x,y)}{\lambda} \in \partial\bar{\varphi}(z_\lambda(x,y)). \tag{5.1}$$

We have $\operatorname{prox}_{\lambda\bar{\varphi}}(x + \lambda v) = x$ for any $v \in \partial\bar{\varphi}(x)$, the prox mapping being 1-Lipschitz, we deduce

$$\|z_\lambda(x,y) - x\| \leq \lambda\|y - v\| \leq 2\lambda L,$$

where we have used Lipschitz continuity of $\tilde{\phi}$ and $\bar{\varphi}$ in a neighborhood of $x \in \operatorname{supp}\tilde{\mu}$, so that $y \in \partial\tilde{\phi}(x)$ and $v \in \partial\bar{\varphi}(x)$ have norm at most L. Now, choose $\lambda = R/(2L)$, we have $\|z_\lambda(x,y) - x\| \leq R$ and $z_\lambda(x,y) \in B(0, 2R)$.

Since $\tilde{\phi}$ is a Brenier potential for the pair $\tilde{\mu}, \tilde{\nu}$, we have for $\tilde{\gamma}$ almost all x, y, $\tilde{\phi}(x) + \tilde{\phi}^*(y) = \langle x, y\rangle$. Using Carlier's inequality in Lemma 4.3 and the definition of z_λ, we obtain

$$\int \bar{\varphi} - \tilde{\phi} d\tilde{\mu} + \int \bar{\varphi}^* - \tilde{\phi}^* d\tilde{\nu} = \int \bar{\varphi}(x) + \bar{\varphi}^*(y) - \langle x, y\rangle\, d\tilde{\gamma}(x,y) \geq \int \frac{1}{\lambda}\|x - z_\lambda(x,y)\|^2 d\tilde{\gamma}(x,y) \tag{5.2}$$

[1] Both works were conducted independently. See also the discussion in introduction and conclusion.

Set $\gamma_0 = T_\sharp \tilde{\gamma}$ and μ_0 its first marginal. We have that $\text{supp } \gamma_0 \subset \text{graph } \partial\bar{\varphi}$ by construction, see (5.1) and $\text{Supp } \mu_0 \subset B(0, 2R)$ by the argument above. We also have $\|T(x,y)-(x,y)\|^2 = (1+1/\lambda^2)\|x-z(x,y)\|^2$ so that

$$W_2^2(\tilde{\gamma}, \gamma_0) \leq \int \|T(x,y)-(x,y)\|^2 d\tilde{\gamma}(x,y) = \left(1+\frac{1}{\lambda^2}\right)\int \|x-z_\lambda(x,y)\|^2 d\tilde{\gamma}(x,y)$$
$$\leq \left(\lambda+\frac{1}{\lambda}\right)\left(\int \bar{\varphi}-\tilde{\phi} d\tilde{\mu} + \int \bar{\varphi}^*-\tilde{\phi}^* d\tilde{\nu}\right),$$

where the last inequality is from (5.2). This is our first inequality.

Let us proceed to the second inequality. Denoting by π_x the projection on the x coordinate, we have that $(\pi_x, z_\lambda)_\sharp \tilde{\gamma}$ has first marginal $\tilde{\mu}$ and second marginal μ_0. Indeed, μ_0 is the first marginal of γ_0, i.e. $\mu_0 = \pi_{x\sharp}\gamma_0 = \pi_{x\sharp}(T_\sharp\tilde{\gamma}) = (\pi_x \circ T)_\sharp\tilde{\gamma} = z_{\lambda\sharp}\tilde{\gamma}$. Similarly, $\pi_{y\sharp}[(\pi_x, z_\lambda)_\sharp\tilde{\gamma}] = [\pi_y \circ (\pi_x, z_\lambda)]_\sharp\tilde{\gamma} = z_{\lambda\sharp}\tilde{\gamma}$ is the second marginal of $(\pi_x, z_\lambda)_\sharp\tilde{\gamma}$. Thus, $(\pi_x, z_\lambda)_\sharp\tilde{\gamma}$ defines a valid coupling for $\tilde{\mu}$ and μ_0 and we deduce our second inequality

$$W_2^2(\tilde{\mu}, \mu_0) \leq \int \|x-z_\lambda(x,y)\|^2 d\tilde{\gamma}(x,y) \leq \lambda\left(\int \bar{\varphi}-\tilde{\phi} d\tilde{\mu} + \int \bar{\varphi}^*-\tilde{\phi}^* d\tilde{\nu}\right)$$

□

The next lemma allows to control the stability of optimal couplings supported on the graph of the same subdifferential, as a function of the variations of the source marginal.

Lemma 5.2. *Fix $R > 0$, $M > 0$. Let μ, β be supported in B_R such that $\mu(A) \leq M\lambda(A)$ for any measurable set A. Assume that $\bar{\varphi}$ is convex lower semicontinuous, L-Lipschitz on B_{2R}. Set $\gamma = (id, \nabla\bar{\varphi})_\sharp\mu$, and assume that θ is a coupling with first marginal β and is supported on the graph of $\partial\bar{\varphi}$. Consider the constant $C_{R,d}$ in Lemma 4.4.*

Case 1: *if $\left(\frac{6W_2^2(\mu,\beta)}{(2MC_{R,d})^{2/3}}\right)^{\frac{3}{8}} > \frac{R}{8}$, then*

$$W_2^2(\gamma,\theta) \leq \left(1+\frac{256L^2}{R^2}\right)W_2^2(\mu,\beta) + L^2MC_{R,d}\left(\frac{6W_2^2(\mu,\beta)}{(2C_{R,d})^{2/3}}\right)^{\frac{3}{8}}.$$

Otherwise:

$$W_2^2(\gamma,\theta) \leq W_2^2(\mu,\beta) + 5L^2(2LMC_{r,d})^{\frac{1}{2}}W_2^2(\mu,\beta)^{\frac{1}{4}}.$$

Proof. Denote by S the optimal mapping from μ to β, see Lemma 2.1. Define γ_x the disintegration of γ, such that for μ almost all x, $\text{supp } \gamma_x \subset \partial\bar{\varphi}(x) = \{\nabla\bar{\varphi}(x)\}$, and for any measurable function on $h\colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$, $\int h(x,y)d\gamma(x,y) = \int h(x,y)d\gamma_x(y)d\mu(x)$. Similarly, θ_z is the disintegration of θ such that $\text{supp } \theta_z \in \partial\bar{\varphi}\,(z)$ for all z and for any measurable function on $h\colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$, $\int h(x,y)d\theta(x,y) = \int h(x,y)d\theta_x(y)d\beta(x)$.

Consider a coupling Γ, between γ and θ as follows, the marginal in x is μ and the remaining variables are independent conditioned on x, with distribution

$$y, \tilde{x}, \tilde{y}|x \sim \gamma_x, S(x), \theta_{S(x)}.$$

We have $S_\sharp \mu = \beta$ so that the second marginal of Γ is θ. The first marginal is obviously γ. The associated quadratic cost provides an upper bound on the W_2 distance between the couplings,

$$W_2^2(\gamma,\theta) \leq \int \|x-\tilde{x}\|^2 + \|y-\tilde{y}\|^2 d\Gamma(x,y,\tilde{x},\tilde{y}) = \int \|x - S(x)\|^2 + \|\nabla\bar{\varphi}(x) - \tilde{y}\|^2 d\theta_{S(x)}(\tilde{y}) d\mu(x)$$

Choose $\eta \in]0, R[$ and $\alpha \geq 0$. Set $\Omega_\eta = \{x | \|x - S(x)\| \geq \eta\}$, by Markov's inequality, we have

$$\mu(\Omega_\eta) \leq \frac{1}{\eta^2} \int \|x - S(x)\|^2 d\mu(x) = \frac{1}{\eta^2} W_2^2(\mu,\beta).$$

We deduce that

$$W_2^2(\gamma,\theta) \leq \left(1 + \frac{4L^2}{\eta^2}\right) W_2^2(\mu,\beta) + \int_{x \notin \Omega_\eta} \|\nabla\bar{\varphi}(x) - \tilde{y}\|^2 d\theta_{S(x)}(\tilde{y}) d\mu(x)$$

where we have used that $\|\nabla\bar{\varphi}(x)\| \leq L$ for μ almost all x, and θ is supported on its graph so that $\|\tilde{y}\| \leq L$ for μ almost all x and $\theta_{S(x)}$ almost all $\tilde{y}$, by Lipschitz continuity of $\bar{\varphi}$.

Set $\Sigma_{\eta,\alpha} = \{x \notin \Omega_\eta | \mathrm{diam}(\partial\bar{\varphi}(B(x,\eta))) \geq \alpha\}$. Note that for all $x \in B_R$: $B(x,\eta) \subset B(0,2R)$ We have, using similar bounds on $\nabla\bar{\varphi}$ and $\tilde{y}$:

$$\begin{aligned}\int_{x \notin \Omega_\eta} \|\nabla\bar{\varphi}(x) - \tilde{y}\|^2 d\theta_{S(x)}(\tilde{y}) d\mu(x) &\leq 4L^2 \mu(\Sigma_{\eta,\alpha}) + \alpha^2 \\ &\leq 4L^2 M \ \mathrm{Vol}(B(0,8\eta))\, \mathcal{N}(\Sigma_{\eta,\alpha} \cap B(0,R), 8\eta) + \alpha^2. \qquad (5.3)\end{aligned}$$

We get for any $\eta > 0$, $\alpha \geq 0$

$$W_2^2(\gamma,\theta) \leq \left(1 + \frac{4L^2}{\eta^2}\right) W_2^2(\mu,\beta) + 4L^2 M \ \mathrm{Vol}(B(0,8\eta))\, \mathcal{N}(\Sigma_{\eta,\alpha} \cap B(0,R), 8\eta) + \alpha^2 \qquad (5.4)$$

We are now going to treat two cases and choose corresponding values for η and α, note that we do not optimize constants. First consider the case $\left(\frac{6W_2^2(\mu,\beta)}{(2MC_{R,d})^{2/3}}\right)^{\frac{3}{8}} > \frac{R}{8}$. We choose $\eta = \frac{R}{8}$, $\alpha = 0$, and obtain using the volume estimates in Lemma A.2 and the definition of $C_{R,d}$ from Lemma 4.4,

$$\begin{aligned}W_2^2(\gamma,\theta) &\leq \left(1 + \frac{256L^2}{R^2}\right) W_2^2(\mu,\beta) + 4L^2 M \mathrm{Vol}(B(0,8\eta)) \\ &\leq \left(1 + \frac{256L^2}{R^2}\right) W_2^2(\mu,\beta) + \frac{1}{8} L^2 M C_{R,d} R \\ &\leq \left(1 + \frac{256L^2}{R^2}\right) W_2^2(\mu,\beta) + L^2 M C_{R,d} \left(\frac{6W_2^2(\mu,\beta)}{(2C_{R,d})^{2/3}}\right)^{\frac{3}{8}}.\end{aligned}$$

Second consider the case $\left(\frac{6W_2^2(\mu,\beta)}{(2MC_{R,d})^{2/3}}\right)^{\frac{3}{8}} \leq \frac{R}{8}$. We get from (5.4) and Lemma 4.1, for any $\eta \leq R/4$ and $\alpha \geq 0$,

$$W_2^2(\gamma,\theta) \leq \left(1 + \frac{4L^2}{\eta^2}\right) W_2^2(\mu,\beta) + \frac{L^3 M C_{R,d} \eta}{\alpha} + \alpha^2.$$

Choosing $\alpha = (L^3 M C_{r,d}\eta)^{\frac{1}{3}}$, we obtain

$$W_2^2(\gamma,\theta) \le \left(1+\frac{4L^2}{\eta^2}\right) W_2^2(\mu,\beta) + 2(L^3 M C_{r,d}\eta)^{\frac{2}{3}}.$$

Choosing $\eta = \left(6\frac{W_2^2(\mu,\beta)}{(2MC_{r,d})^{2/3}}\right)^{\frac{3}{8}} \le \frac{R}{8}$, we obtain

$$\begin{aligned}
&W_2^2(\gamma,\theta) \le \left(1+\frac{4L^2}{\eta^2}\right) W_2^2(\mu,\beta) + 2(L^3 M C_{r,d}\eta)^{\frac{2}{3}} \\
=& W_2^2(\mu,\beta) + 4L^2\left(\frac{1}{6}(2MC_{r,d})^{2/3}\right)^{\frac{3}{4}} W_2^2(\mu,\beta)^{\frac{1}{4}} + 2(L^3 M C_{r,d})^{\frac{2}{3}}\left(\frac{6}{(2MC_{r,d})^{2/3}}\right)^{\frac{1}{4}} W_2^2(\mu,\beta)^{\frac{1}{4}} \\
=& W_2^2(\mu,\beta) + W_2^2(\mu,\beta)^{\frac{1}{4}} L^2 (2MC_{r,d})^{\frac{1}{2}}\left(4\cdot 6^{-3/4} + 2\cdot 6^{1/4}\right) \\
\le& W_2^2(\mu,\beta) + 5L^2(2LMC_{r,d})^{\frac{1}{2}} W_2^2(\mu,\beta)^{\frac{1}{4}}.
\end{aligned}$$

□

5.2 Stability of optimal couplings under variations of source and target

The following is the main stability result of this section. A simpler corollary is given right after. The main idea of the proof is illustrated in Figure 2.

Theorem 5.3. *For any d, R_μ, R_ν, M, there is $C > 0$ such that the following holds. Let μ and ν be probability measures supported on B_{R_μ} and B_{R_ν} respectively, with $\mu(A) \le M\lambda(A)$ for any measurable set $A \subset \mathbb{R}^d$. Let $\bar\varphi \in \mathcal{C}_{4R_\mu,4R_\nu}$ be an optimal Brenier potential (as in Lemma 2.2) for these measures and γ be their optimal coupling. Let $\tilde\mu$, $\tilde\nu$ be probability measures also supported on B_{R_μ} and B_{R_ν} respectively with associated Brenier potential $\tilde\phi \in \mathcal{C}_{4R_\mu,4R_\nu}$ and coupling $\tilde\gamma$. Then*

$$\frac{1}{C^4} W_2(\gamma,\tilde\gamma)^4 \le W_2(\mu,\tilde\mu) + \left(\int \bar\varphi - \tilde\phi d\tilde\mu + \int \bar\varphi^* - \tilde\phi^* d\tilde\nu\right)^{\frac{1}{2}}$$

Proof. We start with Lemma 5.1 which applies under the hypotheses of Lemma 5.3 with the same notation with $R = R_\mu$ and $L = 2R_\nu$. We get γ_0 supported on $\partial\bar\varphi$ with first marginal μ_0 supported on B_{2R_μ}. Since both γ_0 and γ are supported on $\partial\bar\varphi$ with marginals in B_{2R_μ} on which $\bar\varphi$ is Lipschitz, Lemma 5.2 applies with $\theta = \gamma_0$, $R = 2R_\mu$ and $L = 4R_\nu$.

All Wasserstein distances are bounded by constants depending on d, M, R_μ, R_ν. Similarly as in the proof of Lemma 4.7, it is possible to obtain $C > 0$ by adjusting the constants in Lemma 5.2 so that only the smallest power dominates in all cases, resulting in $W_2(\gamma_0,\gamma) \le C W_2(\mu_0,\mu)^{\frac{1}{4}}$. Note again that both γ_0 and γ are supported on $\partial\bar\varphi$ and their first marginal, μ_0 and μ are supported on B_{2R_μ} on which $\bar\varphi$ is assumed to be Lipschitz. We have

$$\begin{aligned}
W_2(\tilde\gamma,\gamma) &\le W_2(\tilde\gamma,\gamma_0) + W_2(\gamma_0,\gamma) \\
&\le W_2(\tilde\gamma,\gamma_0) + C W_2(\mu_0,\mu)^{\frac{1}{4}} \le W_2(\tilde\gamma,\gamma_0) + C(W_2(\mu_0,\tilde\mu) + W_2(\mu,\tilde\mu))^{\frac{1}{4}}.
\end{aligned}$$

Set $\delta = \int \bar{\varphi} - \tilde{\phi} d\tilde{\mu} + \int \bar{\varphi}^* - \tilde{\phi}^* d\tilde{\nu}$. Considering Lemma 5.1, we may adjust the constant C such that

$$W_2(\tilde{\gamma}, \gamma) \leq C\sqrt{\delta} + C\left(W_2(\mu, \tilde{\mu}) + \sqrt{\delta}\right)^{\frac{1}{4}}$$

Similarly to the proof of Lemma 4.7, δ is bounded by a constant depending only on d, M, R_μ, R_ν. We may thus keep the smallest power and adjust the constant C to get the desired estimate. □

Corollary 5.4. *For any d, R_μ, R_ν, M, there is $C > 0$ such that for any probability distribution μ, ν supported on B_{R_μ} and B_{R_ν} respectively, with $\mu(A) \leq M\lambda(A)$ and optimal coupling γ, for any probability distribution $\tilde{\mu}$, $\tilde{\nu}$ also supported on B_{R_μ} and B_{R_ν} respectively, with optimal coupling $\tilde{\gamma}$, we have*

$$\frac{1}{C^8} W_2(\gamma, \tilde{\gamma})^8 \leq W_2(\mu, \tilde{\mu}) + W_2(\nu, \tilde{\nu}).$$

Proof. By Lemma 2.2 there is an optimal Brenier potential $\bar{\varphi} \in \mathcal{C}_{4R_\mu, 4R_\nu}$, for measures μ, ν. The same is true for measures $\tilde{\mu}$, $\tilde{\nu}$ and Brenier potential $\tilde{\phi} \in \mathcal{C}_{4R_\mu, 4R_\nu}$. The hypotheses of Lemma 5.3 are satisfied for $\bar{\varphi}$ and $\tilde{\phi}$. Furthermore using Lemma 2.1 for measures μ and ν, we have

$$\int_{\mathbb{R}^d} (\bar{\varphi} - \tilde{\phi}) d\mu + \int_{\mathbb{R}^d} (\bar{\varphi}^* - \tilde{\phi}^*) d\nu \leq 0.$$

From Lemma 5.3 there is $C > 0$, such that for any $\mu, \nu, \tilde{\mu}, \tilde{\nu}$ as in the statement, we have

$$\begin{aligned} \frac{1}{C^4} W_2(\gamma, \tilde{\gamma})^4 \leq\ & W_2(\mu, \tilde{\mu}) + \left(\int \bar{\varphi} - \tilde{\phi} d\tilde{\mu} + \int \bar{\varphi}^* - \tilde{\phi}^* d\tilde{\nu} \right)^{\frac{1}{2}} \\ \leq\ & W_2(\mu, \tilde{\mu}) + \left(\int \bar{\varphi} - \tilde{\phi} d(\tilde{\mu} - \mu) + \int \bar{\varphi}^* - \tilde{\phi}^* d(\tilde{\nu} - \nu) \right)^{\frac{1}{2}} \end{aligned} \tag{5.5}$$

In view of applying [Villani, 2021, Theorem 1.14], we remark that $\bar{\varphi} - \tilde{\phi}$ and $\bar{\varphi}^* - \tilde{\phi}^*$ are $8R_\nu$ and $8R_\mu$ Lipschitz respectively. Therefore

$$\begin{aligned} \int \bar{\varphi} - \tilde{\phi} d(\tilde{\mu} - \mu) + \int \bar{\varphi}^* - \tilde{\phi}^* d(\tilde{\nu} - \nu) &\leq 8R_\nu W_1(\tilde{\mu}, \mu) + 8R_\mu W_1(\tilde{\nu}, \nu) \\ &\leq 8 \max(R_\mu, R_\nu)(W_2(\tilde{\mu}, \mu) + W_2(\tilde{\nu}, \nu)). \end{aligned} \tag{5.6}$$

Combining (5.5) and (5.6) we obtain

$$\begin{aligned} \frac{1}{C^4} W_2(\gamma, \tilde{\gamma})^4 &\leq W_2(\mu, \tilde{\mu}) + 3\sqrt{\max(R_\mu, R_\nu)}(W_2(\tilde{\mu}, \mu) + W_2(\tilde{\nu}, \nu))^{\frac{1}{2}} \\ &\leq W_2(\mu, \tilde{\mu}) + 3\sqrt{\max(R_\mu, R_\nu)} \left(W_2(\tilde{\mu}, \mu)^{\frac{1}{2}} + W_2(\tilde{\nu}, \nu)^{\frac{1}{2}} \right). \end{aligned}$$

With a similar reasoning as the one made in the proof of Lemma 4.7 and Lemma 5.3 we can adjust the constants in the previous equation so that only the smallest powers dominate. This concludes the proof. □

Remark 5.5. *The formula above implies uniqueness of the optimal coupling between μ and ν (take $\tilde{\mu} = \mu$, $\tilde{\nu} = \nu$). This coincides with the absolute continuity assumption, which is sufficient for uniqueness.*

6 Conclusion

We have proposed an estimator for Monge maps, it is given by the gradient of a convex function and enjoys a closed form expression based on discrete dual solutions. Moreover, we derived finite sample guaranties leveraging a dual error bound for Brenier dual potentials, for both the general and semi-discrete cases. Up to powers, these bounds match existing results in the literature under considerably stronger assumptions. Our results hold in a very general and explicit setting: compact support and upper bounded source density. This allows to consider, in particular, possibly discontinuous Brenier potentials as well as source measures with disconnected support.

Two questions remain. First, regarding the optimality of our finite sample bounds: could they be improved under our general assumptions or are there matching lower bounds? Second, regarding the adaptability of our estimator: under the Monge diffeomorphism assumption described in Section 3.3, does it achieve existing minimax rates? These questions are left to future work.

On exponent improvements: After the diffusion of the first version of this work, we became aware of [Mérigot, 2026] which also combines results from [Carlier et al., 2025] and convex analysis results in [Carlier, 2023] to obtain various stability bounds in optimal transport. In particular, the bound in Lemma 4.9 is described with an exponent $1/4$, which is optimal [Letrouit, 2025], instead of $1/5$ in our work. Technically, the difference in exponents comes from the replacement of pointwise estimate in Lemma 4.1 with a form integrated over α [Carlier et al., 2025, Corollary 2.2]. We neglected this aspect in the first version of this work and a slight technical modification of our arguments yields improved statistical estimation rates. We mention these rates here, but otherwise leave the main text unchanged in its original form in order to preserve the independence of our contribution and that of [Mérigot, 2026]. More details are given in Section A.4.

- Lemma 3.7 holds with exponent arbitrarily close to $1/3$ instead of $1/4$.
- Lemma 3.14 holds with exponent arbitrarily close to $1/6$ instead of $1/8$.
- Lemma 3.17 holds with exponent $1/6$ instead of $1/8$.

A Appendix

A.1 Technical Lemmas

Lemma A.1. *For all $\varphi_0 \in \Gamma_0(\mathbb{R}^d)$ which is R_ν-Lipschitz on B_{R_μ} there exists $\varphi_1 \in \mathcal{C}_{R_\mu,R_\nu}$ such that $\varphi_1 = \varphi_0$ over B_{R_μ}, and $\varphi_0^* = \varphi_1^*$ on $\partial\varphi_0(B_{R_\mu})$.*

Proof. Define $\varphi_1 := \varphi_0 + \delta(.|\bar{B}_{R_\mu})$ where $\delta(.|C)$ denotes the indicator function which values 0 in C and $+\infty$ elsewhere. By construction, we have that $\varphi_1 \in \Gamma_0(\mathbb{R}^d)$ and is R_ν-Lipschitz on B_{R_μ}. It also results classically that φ_1^* is globally R_μ-Lipschitz. Indeed, φ_1^* is finite everywhere as it is defined by maximization of an upper semicontinuous function on a compact set. Therefore, $\partial\varphi_1^*$ has full domain and $x \in \partial\varphi_1^*(y)$ implies $y \in \partial\varphi_1(x)$ ([Rockafellar, 1970, Theorem 23.5]) implying that $\varphi_1(x)$ is finite . By construction of φ_1 this means that $\|x\| \leq R_\mu$ and by [Rockafellar, 1970, Theorem 24.7], φ_1^* is R_μ-Lipschitz.

For the last part, since $\varphi_0 = \varphi_1$ on B_{R_μ}, it holds that $\partial\varphi_0 = \partial\varphi_1$ on B_{R_μ}. For any $y \in \partial\varphi_0(B_{R_\mu})$, there is $x \in B_{R_\mu}$ such that $y \in \partial\varphi_0(x)$. Since we have $\varphi_1(x) = \varphi_0(x)$ and also $y \in \partial\varphi_1(x)$, it holds that $\varphi_1^*(y) = \langle x, y\rangle - \varphi_1(x) = \langle x, y\rangle - \varphi_0(x) = \varphi_0^*(y)$ using the equality case of Fenchel-Young's inequality [Rockafellar, 1970, Theorem 23.5]. □

Lemma A.2. *For all $r > 0$ we have the following upper bounds on the volume of balls of radius r:*

$$\mathrm{Vol}(B(0,r)) \leq \frac{e}{\sqrt{2\pi}\sqrt{\frac{d}{2}+1}} \left(\frac{\pi e}{\left(\frac{d}{2}+1\right)} \right)^{\frac{d}{2}} r^d$$

$$\mathrm{Vol}(B(0,8r)) \leq \frac{e}{\sqrt{2\pi}\sqrt{\frac{d}{2}+1}} \left(\frac{\pi e 64}{\left(\frac{d}{2}+1\right)} \right)^{\frac{d}{2}} r^d.$$

Proof. The Gamma function admits the following lower bound, given for example in [Olver et al., 2010, Section 5.6]: for all $z > 0$, $\Gamma(z) \geq \sqrt{2\pi} z^{z-1/2} e^{-z}$. Combining this bound with the expression of the volume of B_r: $\mathrm{Vol}(B_r) = \frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2}+1\right)} r^d$ we obtain:

$$\mathrm{Vol}(B_r) \leq \frac{\pi^{d/2}}{\sqrt{2\pi}\left(\frac{d}{2}+1\right)^{\frac{d+1}{2}} e^{-\left(\frac{d}{2}+1\right)}} r^d = \frac{e}{\sqrt{2\pi}\sqrt{\frac{d}{2}+1}} \left(\frac{\pi e}{\left(\frac{d}{2}+1\right)} \right)^{\frac{d}{2}} r^d.$$

We deduce the bound on $\mathrm{Vol}(B(0,8r))$. □

A.2 Proof of the main duality theorem

Proof of Lemma 2.1. This is mostly classical, we provide arguments and literature pointers for completeness. [Santambrogio, 2015, Theorem 1.40] ensures that there exists Kantorovich dual potentials optimal for the following problem

$$\max_{(f,g) \in L^1(\mu) \times L^1(\nu)} \int_{\mathbb{R}^d} f d\mu + \int_{\mathbb{R}^d} g d\nu. \tag{A.1}$$

Moreover the functions (f, g) can be chosen such that $\varphi := \frac{\|.\|^2}{2} - f$ and $\psi := \frac{\|.\|^2}{2} - g$ verify $\varphi^* = \psi$ and $\psi^* = \varphi$ and thus φ and ψ are in $\Gamma_0(\mathbb{R}^d)$. Furthermore, [Santambrogio, 2015, Theorem 1.40] provides equality between (A.1) and (PP). Therefore using this change of variables we obtain (PP) = (DP) and both are being attained.

We now prove the equivalence between the three characterizations. The first equivalence ($1 \iff 2$) is a consequence of (PP)=(DP). Let us show ($2 \iff 3$). Expanding the square in $\int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{1}{2}\|x-y\|^2 d\gamma(x,y)$ and using the fact that γ has marginals μ and ν shows that point 2 is equivalent to $\int_{\mathbb{R}^d \times \mathbb{R}^d} [\varphi(x) + \varphi^*(y) - \langle x, y\rangle] d\gamma(x,y) = 0$. By the Fenchel-Young inequality the integrand is positive, and using Lemma A.5 this is equivalent to $\operatorname{supp}\gamma \subset \{(x,y),\ \varphi(x) + \varphi^*(y) = \langle x, y\rangle\} = \operatorname{graph}(\partial\varphi)$.

Let us now prove that an optimal potential in (DP) may be chosen Lipschitz. Let φ be an optimal dual potential in (DP) and consider $\varphi_1 = (\varphi^* + \delta(.|\bar{B}_{R_\nu}))^*$. Classically (see [Rockafellar, 1970, Theorems 23.5 and 24.7]), φ_1 is R_ν-Lipschitz. Let us show that φ_1 is also optimal for (DP). Fix $(x,y) \in \operatorname{supp}\gamma$, by the third characterization, we have $y \in \partial\varphi(x)$, hence $y \in \operatorname{argmax}_{z \in \mathbb{R}^d} \langle x, z\rangle - \varphi^*(z)$ by [Rockafellar, 1970, Theorem 23.5], so that

$$\varphi(x) = \varphi^{**}(x) = \sup_z \langle x, z\rangle - \varphi^*(z) = \langle x, y\rangle - \varphi^*(y) = \max_{\|z\| \leq R} \langle x, z\rangle - \varphi^*(z) = \varphi_1(x).$$

Also we obviously have $\varphi_1^*(y) = \varphi^*(y)$ for all $y \in \operatorname{supp} \nu$. Since γ is a coupling with marginals μ and ν the optimality of φ_1 is deduced from the previous arguments by integration:

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} [\varphi_1(x) + \varphi_1^*(y)] d\gamma(x,y) = \int_{\mathbb{R}^d \times \mathbb{R}^d} [\varphi(x) + \varphi^*(y)] d\gamma(x,y).$$

Suppose now that μ is absolutely continuous. The existence of a transport map can be deduced from Rademacher's Theorem [Santambrogio, 2015, Theorem 1.17]. Any Lipschitz optimal Brenier potential is differentiable almost everywhere such that $\partial\varphi(x) = \{\nabla\varphi(x)\}$ for μ almost all x. Consequently, since γ has first marginal μ and is supported on $\operatorname{graph} \partial\varphi$, for any measurable function f : $\int f(x,y)\, d\gamma(x,y) = \int f\big(x, \nabla\varphi(x)\big)\, d\mu(x)$.

Uniqueness of the map can be shown using [Santambrogio, 2015, Remark 1.19] for example. □

A.3 Proof illustration

Figure 2 illustrates the proof of Lemma 5.3.

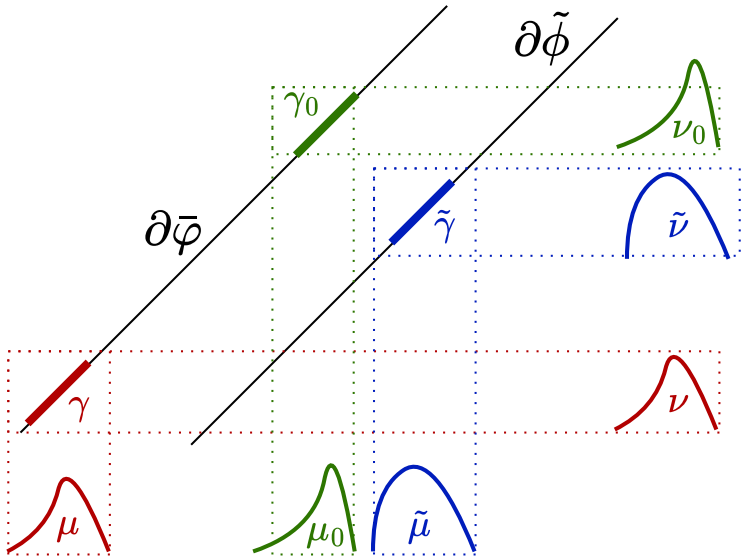

Figure 2: A visual illustration of the proof of Lemma 5.3. γ and $\tilde{\gamma}$ are supported on the graph of $\partial\bar{\varphi}$ and $\partial\tilde{\phi}$ respectively. Lemma 5.1 provides γ_0, on the graph of $\partial\bar{\varphi}$, such that $W_2(\tilde{\gamma}, \gamma_0)$ is controlled by a suboptimality gap. Since γ and γ_0 are both supported on the graph of $\partial\bar{\varphi}$, Lemma 5.2 allows to control $W_2(\gamma, \gamma_0)$ with a $W_2(\mu, \mu_0)$ term, in turn controlled by $W_2(\mu, \tilde{\mu})$ and the suboptimality gap in Lemma 5.1.

A.4 On obtaining improved rates

As described in the conclusion of the main text, our statistical convergence rates could be improved, see also [Mérigot, 2026]. The main argument is as follows, it relies on [Carlier et al., 2025, Corollary 2.2] instead of Lemma 4.1. In the proof of Lemma 4.4, equation (4.4) could be replaced by the following, for $q > 1$:

$$\int \|v_\lambda - \nabla\phi\|^q d\mu \leq \int \operatorname{diam}(\partial\phi(B(x,\eta)))^q d\mu(x) \leq M \int \operatorname{diam}(\partial\phi(B(x,\eta)))^q dx \leq ML^qC\eta$$

for some constant C given in [Carlier et al., 2025, Corollary 2.2]. The obtained bound can then be combined with a computation similar to Lemma 4.6 to obtain case 2 in Lemma 4.4, with an exponent $q+2$ instead of $q+3$ for $q>1$.

Similarly in the proof of Lemma 5.2, equation (5.3) could be replaced by

$$\int_{x\notin\Omega_\eta} \|\nabla\bar{\varphi}(x)-\tilde{y}\|^2 d\theta_{S(x)}(\tilde{y})d\mu(x) \leq M\int \operatorname{diam}(\partial\phi(B(x,\eta)))^2 dx \leq ML^2C\eta$$

for some constant C given in [Carlier et al., 2025, Corollary 2.2]. Similar computations lead to an inequality of the form

$$W_2^2(\gamma,\theta) \leq \left(1+\frac{4L^2}{\eta^2}\right) W_2^2(\mu,\beta) + c\eta$$

for some constant c. Optimizing over η leads to an exponent $1/3$ instead of $1/4$ in case two of Lemma 5.2. The same reasoning as in Lemma 5.3 leads to an exponent $1/6$ instead of $1/8$.

A.5 Measure theoretical lemmas

The following are classical, we provide proof arguments for completeness.

Definition A.3. *The support of a probability measure μ on $\mathbb{R}^d$ is the smallest closed set $X\subset\mathbb{R}^d$ such that $\mu(X)=1$. That is:*

$$\operatorname{supp}\mu = \bigcap\left\{X\subset\mathbb{R}^d \mid X \text{ closed},\ \mu(X)=1\right\}.$$

Or equivalently:

$$\operatorname{supp}\mu = \{x\in\mathbb{R}^d \mid \mu(B(x,\epsilon))>0\ \forall\epsilon>0\}.$$

Definition A.4. *For $i=1,2$ we denote $\pi_i : \mathbb{R}^d\times\mathbb{R}^d \longrightarrow \mathbb{R}^d$ the projection mapping on the first and respectively second component, i.e for every $(x_1,x_2)\in\mathbb{R}^d\times\mathbb{R}^d$: $\pi_i(x_1,x_2)=x_i$.*

Proposition A.5. *Let μ be a probability measure and f be a non-negative lower semi-continuous function. Then $\int_{\mathbb{R}^d} f d\mu = 0$ if and only if $\operatorname{supp}\mu\subset\{x\in\mathbb{R}^d,\ f(x)=0\}$.*

Proof. Let $x\in\operatorname{supp}\mu$. For all $\epsilon>0$ we always have $\int_{B(x,\epsilon)} f d\mu = 0$. On the other hand we also have $\int_{B(x,\epsilon)} f d\mu \geq \inf_{t\in B(x,\epsilon)} f(t)\mu(B(x,\epsilon))$. By definition of the support $\mu(B(x,\epsilon))>0$ for all $\epsilon>0$. It results that for any $x\in\operatorname{supp}\mu$ and for any $\epsilon>0$, $\inf_{t\in B(x,\epsilon)} f(t)=0$. Taking the limit as $\epsilon\longrightarrow 0$ we thus obtain $\lim_{\epsilon\to 0}\inf_{t\in B(x,\epsilon)} f(t) = \liminf_{t\to x} f(t) \geq f(x)$ by lower semi-continuity. This shows that $f(x)=0$. □

Lemma A.6. *Let $\mu,\ \nu$ be be compactly supported probability measures on $\mathbb{R}^d$ and $\gamma\in\Pi(\mu,\nu)$. Then*

$$\operatorname{supp}\mu = \{x\in\mathbb{R}^d,\ \exists y\in\operatorname{supp}\nu \text{ s.t } (x,y)\in\operatorname{supp}(\gamma)\}.$$

Proof. By [McCann and Trokhimtchouk, 2010, Lemma 3] we have:

$$\operatorname{supp}\mu = \overline{\operatorname{proj}_1(\operatorname{supp}\gamma)} \tag{A.2}$$

where $\mathrm{proj}_i : \mathbb{R}^d \times \mathbb{R}^d \longrightarrow \mathbb{R}^d$ denotes the canonical projection mapping on the i'th component. We thus deduce the following sequence of inclusions.

$$\operatorname{supp}\gamma \ \subset \mathrm{proj}_1(\operatorname{supp}\gamma) \times \mathrm{proj}_2(\operatorname{supp}\gamma) \subset \overline{\mathrm{proj}_1(\operatorname{supp}\gamma)} \times \overline{\mathrm{proj}_2(\operatorname{supp}\gamma)} = \operatorname{supp}\mu \times \operatorname{supp}\nu.$$

We deduce that $\operatorname{supp}\gamma$ is compact, and for any $(x, y) \in \operatorname{supp}\gamma$, $y \in \operatorname{supp}\nu$. It follows that

$$\begin{aligned} \mathrm{proj}_1(\operatorname{supp}\gamma) &:= \left\{ x \in \mathbb{R}^d,\ \exists y \in \mathbb{R}^d,\ (x, y) \in \operatorname{supp}\gamma \right\} \\ &= \left\{ x \in \mathbb{R}^d,\ \exists y \in \operatorname{supp}\nu,\ (x, y) \in \operatorname{supp}\gamma \right\}. \end{aligned} \tag{A.3}$$

It remains to show that $\mathrm{proj}_1(\operatorname{supp}\gamma)$ is closed, which follows from boundedness. Let $(x_n)_{n\in\mathbb{N}^*} \subset \mathrm{proj}_1(\operatorname{supp}\gamma)$ be a convergent sequence. For every $n \in \mathbb{N}^*$, there exists $y_n \in \operatorname{supp}\nu$ such that $(x_n, y_n) \in \operatorname{supp}\gamma$. Since $\operatorname{supp}\gamma$ is compact, there exists a subsequence of $(x_n, y_n)_n$ converging to some $(x^*, y^*) \in \operatorname{supp}\gamma$. Since $(x_n)_n$ converges, we necessarily have $x^* \in \mathrm{proj}_1(\operatorname{supp}\gamma)$. Hence, $\mathrm{proj}_1(\operatorname{supp}\gamma)$ is closed. Combining (A.2) and (A.3) thus yields the result. □

A.6 Further technical lemmas

Lemma A.7. *Assume $\delta, L, M, C > 0$ and $q \in [1, 2]$. For $\eta = \left(\frac{\sqrt{\delta}}{\sqrt{2L}(MC)^{\frac{1}{q+1}}} \right)^{\frac{2q+2}{q+3}}$, we have*

$$2L(MC\eta)^{\frac{1}{q+1}} + \sqrt{\frac{2L\delta}{\eta}} = 2(2L)^{\frac{q+2}{q+3}}(MC\delta)^{\frac{1}{q+3}}$$

Proof. We compute powers, we have $2L(MC\eta)^{\frac{1}{q+1}} = (2L)^a (MC)^b \delta^c$ with

$$\begin{aligned} a &= 1 - \frac{2q+2}{2(q+1)(q+3)} = \frac{q+2}{q+3} \\ b &= \frac{1}{q+1}\left(1 - \frac{2q+2}{(q+1)(q+3)}\right) = \frac{1}{q+3} \qquad\qquad c = \frac{2q+2}{2(q+1)(q+3)} = \frac{1}{q+3}. \end{aligned}$$

Similarly, we have $\sqrt{\frac{2L\delta}{\eta}} = (2L)^a (MC)^b \delta^c$ with

$$\begin{aligned} a &= \frac{1}{2} + \frac{2q+2}{4(q+3)} = \frac{q+3+q+1}{2(q+3)} = \frac{q+2}{q+3} \\ b &= \frac{2q+2}{2(q+1)(q+3)} = \frac{1}{q+3} \qquad\qquad c = \frac{1}{2} - \frac{2q+2}{4(q+3)} = \frac{q+3-(q+1)}{2(q+3)} = \frac{1}{q+3} \end{aligned}$$

□

Acknowledgments

This work benefited from financial support from the French government managed by the National Agency for Research under the France 2030 program, with the references "ANR-23-PEIA-0004" and "ANR-25-PEIA-0002". EP thanks TSE-P and acknowledges the support of the AI Interdisciplinary Institute ANITI funding, through the ANR under the France 2030 program (grant ANR-23-IACL-0002), Chair TRIAL, ANR MAD and ANR Regulia.

References

[Acosta and Durán, 2004] Acosta, G. and Durán, R. (2004). An optimal poincaré inequality in l^1 for convex domains. *Proceedings of the american mathematical society*, 132(1):195–202.

[Aurenhammer, 1987] Aurenhammer, F. (1987). Power diagrams: properties, algorithms and applications. *SIAM Journal on Computing*, 16(1):78–96.

[Aurenhammer et al., 1998] Aurenhammer, F., Hoffmann, F., and Aronov, B. (1998). Minkowski-type theorems and least-squares clustering. *Algorithmica*, 20(1):61–76.

[Bakry et al., 2008] Bakry, D., Barthe, F., Cattiaux, P., and Guillin, A. (2008). A simple proof of the Poincaré inequality for a large class of probability measures. *Electronic Communications in Probability*, 13:60–66.

[Balakrishnan et al., 2025] Balakrishnan, S., Manole, T., and Wasserman, L. (2025). Statistical inference for optimal transport maps: recent advances and perspectives. *arXiv preprint:2506.19025*.

[Bauschke et al., 2022] Bauschke, H. H., Singh, S., and Wang, X. (2022). On carlier's inequality. *arXiv preprint arXiv:2206.14872*.

[Brenier, 1991] Brenier, Y. (1991). Polar factorization and monotone rearrangement of vector-valued functions. *Communications on pure and applied mathematics*, 44(4):375–417.

[Burachik and Martínez-Legaz, 2025] Burachik, R. S. and Martínez-Legaz, J. (2025). A note on carlier's inequality. *Optimization*, pages 1–6.

[Caffarelli, 1992a] Caffarelli, L. A. (1992a). Boundary regularity of maps with convex potentials. *Communications on pure and applied mathematics*, 45(9):1141–1151.

[Caffarelli, 1992b] Caffarelli, L. A. (1992b). The regularity of mappings with a convex potential. *Journal of the American Mathematical Society*, 5(1):99–104.

[Caffarelli, 1996] Caffarelli, L. A. (1996). Boundary regularity of maps with convex potentials–II. *Annals of mathematics*, 144(3):453–496.

[Carlier, 2023] Carlier, G. (2023). Fenchel–young inequality with a remainder and applications to convex duality and optimal transport. *SIAM Journal on Optimization*, 33(3):1463–1472.

[Carlier et al., 2025] Carlier, G., Delalande, A., and Mérigot, Q. (2025). Quantitative stability of the push-forward operation by an optimal transport map. *Foundations of Computational Mathematics*, 25(4):1259–1286.

[Chewi et al., 2025] Chewi, S., Niles-Weed, J., and Rigollet, P. (2025). Estimation of transport maps. In *Statistical Optimal Transport: École d'Été de Probabilités de Saint-Flour XLIX–2019*, pages 77–94. Springer.

[Chhaibi et al., 2025] Chhaibi, R., Gratton, S., and Vaiter, S. (2025). Faster computation of entropic optimal transport via stable low frequency modes. *arXiv preprint:2506.14780*.

[Cobzas and Mustata, 1978] Cobzas, S. and Mustata, C. (1978). Norm preserving extension of convex Lipschitz functions. *J. Approx. theory*, 24(3):236–244.

[Combettes and Mayrand, 2026] Combettes, P. L. and Mayrand, J. N. (2026). Lower bounds on the haraux function. *Journal of Approximation Theory*, page 106284.

[Cominetti and Martín, 1994] Cominetti, R. and Martín, J. S. (1994). Asymptotic analysis of the exponential penalty trajectory in linear programming. *Mathematical Programming*, 67(1):169–187.

[Cuturi, 2013] Cuturi, M. (2013). Sinkhorn distances: lightspeed computation of optimal transport. *Advances in neural information processing systems*, 26.

[De Lara et al., 2021] De Lara, L., González-Sanz, A., and Loubes, J.-M. (2021). A consistent extension of discrete optimal transport maps for machine learning applications. *arXiv preprint:2102.08644*.

[Del Barrio et al., 2024] Del Barrio, E., González Sanz, A., and Loubes, J.-M. (2024). Central limit theorems for semi-discrete Wasserstein distances. *Bernoulli*, 30(1):554–580.

[Delalande and Mérigot, 2023] Delalande, A. and Mérigot, Q. (2023). Quantitative stability of optimal transport maps under variations of the target measure. *Duke Mathematical Journal*, 172(17):3321–3357.

[Divol et al., 2025] Divol, V., Niles-Weed, J., and Pooladian, A.-A. (2025). Optimal transport map estimation in general function spaces. *The Annals of Statistics*, 53(3):963–988.

[Flamary et al., 2021] Flamary, R., Courty, N., Gramfort, A., Alaya, M. Z., Boisbunon, A., Chambon, S., Chapel, L., Corenflos, A., Fatras, K., Fournier, N., et al. (2021). POT: Python optimal transport. *Journal of Machine Learning Research*, 22(78):1–8.

[Fournier, 2023] Fournier, N. (2023). Convergence of the empirical measure in expected Wasserstein distance: non-asymptotic explicit bounds in $\mathbb{R}^d$. *ESAIM: Probability and Statistics*, 27:749–775.

[Fournier and Guillin, 2015] Fournier, N. and Guillin, A. (2015). On the rate of convergence in Wasserstein distance of the empirical measure. *Probability theory and related fields*, 162(3):707–738.

[Genevay et al., 2016] Genevay, A., Cuturi, M., Peyré, G., and Bach, F. (2016). Stochastic optimization for large-scale optimal transport. *Advances in neural information processing systems*, 29.

[Gigli, 2011] Gigli, N. (2011). On Hölder continuity-in-time of the optimal transport map towards measures along a curve. *Proceedings of the Edinburgh Mathematical Society*, 54(2):401–409.

[Giné and Nickl, 2021] Giné, E. and Nickl, R. (2021). *Mathematical foundations of infinite-dimensional statistical models*. Cambridge university press.

[Gu et al., 2013] Gu, X., Luo, F., Sun, J., and Yau, S.-T. (2013). Variational principles for Minkowski type problems, discrete optimal transport, and discrete Monge-Ampere equations. *arXiv preprint:1302.5472*.

[Gunsilius, 2022] Gunsilius, F. (2022). On the convergence rate of potentials of Brenier maps. *Econometric Theory*, 38(2):381–417.

[Hallin et al., 2021] Hallin, M., del Barrio, E., Cuesta-Albertos, J., and Matrán, C. (2021). Distribution and quantile functions, ranks and signs in dimension d. *The Annals of Statistics*, 49(2):1139–1165.

[Hütter and Rigollet, 2021] Hütter, J.-C. and Rigollet, P. (2021). Minimax estimation of smooth optimal transport maps. *The Annals of Statistics*, 49.

[Lei et al., 2019] Lei, N., Su, K., Cui, L., Yau, S.-T., and Gu, X. D. (2019). A geometric view of optimal transportation and generative model. *Computer Aided Geometric Design*, 68:1–21.

[Letrouit, 2025] Letrouit, C. (2025). Unstable optimal transport maps. *arXiv preprint arXiv:2510.13265*.

[Letrouit and Mérigot, 2024] Letrouit, C. and Mérigot, Q. (2024). Gluing methods for quantitative stability of optimal transport maps. *arXiv preprint arXiv:2411.04908*.

[Lévy, 2024] Lévy, B. (2024). Large-scale semi-discrete optimal transport with distributed Voronoi diagrams. *arXiv preprint:2406.04192*.

[Li and Nochetto, 2021] Li, W. and Nochetto, R. H. (2021). Quantitative stability and error estimates for optimal transport plans. *IMA Journal of Numerical Analysis*, 41(3):1941–1965.

[Makkuva et al., 2020] Makkuva, A., Taghvaei, A., Oh, S., and Lee, J. (2020). Optimal transport mapping via input convex neural networks. In *International Conference on Machine Learning*, pages 6672–6681. PMLR.

[Manole et al., 2024] Manole, T., Balakrishnan, S., Niles-Weed, J., and Wasserman, L. (2024). Plugin estimation of smooth optimal transport maps. *The Annals of Statistics*, 52(3):966–998.

[McCann, 2001] McCann, R. J. (2001). Polar factorization of maps on Riemannian manifolds. *Geometric & Functional Analysis GAFA*, 11(3):589–608.

[McCann and Trokhimtchouk, 2010] McCann, R. J. and Trokhimtchouk, M. (2010). Optimal partition of a large labor force into working pairs. *Economic theory*, 42(2):375–395.

[Mérigot, 2026] Mérigot, Q. (2026). Sharp stability of brenier maps via quantitative regularity of potentials. Technical report.

[Mérigot et al., 2020] Mérigot, Q., Delalande, A., and Chazal, F. (2020). Quantitative stability of optimal transport maps and linearization of the 2-Wasserstein space. In *International Conference on Artificial Intelligence and Statistics*, pages 3186–3196. PMLR.

[Moreau, 1965] Moreau, J.-J. (1965). Proximité et dualité dans un espace hilbertien. *Bulletin de la Société mathématique de France*, 93:273–299.

[Muzellec et al., 2021] Muzellec, B., Vacher, A., Bach, F., Vialard, F.-X., and Rudi, A. (2021). Near-optimal estimation of smooth transport maps with kernel sums-of-squares. *arXiv preprint arXiv:2112.01907*.

[Olver et al., 2010] Olver, F. W., Lozier, D. W., Boisvert, R. F., and Clark, C. W. (2010). *NIST handbook of mathematical functions hardback and CD-ROM*. Cambridge university press.

[Paty et al., 2020] Paty, F.-P., d'Aspremont, A., and Cuturi, M. (2020). Regularity as regularization: smooth and strongly convex Brenier potentials in optimal transport. In *International Conference on Artificial Intelligence and Statistics*, pages 1222–1232. PMLR.

[Perrot et al., 2016] Perrot, M., Courty, N., Flamary, R., and Habrard, A. (2016). Mapping estimation for discrete optimal transport. *Advances in Neural Information Processing Systems*, 29.

[Peyré and Cuturi, 2019] Peyré, G. and Cuturi, M. (2019). Computational optimal transport: with applications to data science. *Foundations and Trends® in Machine Learning*, 11(5-6):355–607.

[Pooladian et al., 2023] Pooladian, A.-A., Divol, V., and Niles-Weed, J. (2023). Minimax estimation of discontinuous optimal transport maps: The semi-discrete case. In *International Conference on Machine Learning*, pages 28128–28150. PMLR.

[Pooladian and Niles-Weed, 2021] Pooladian, A.-A. and Niles-Weed, J. (2021). Entropic estimation of optimal transport maps. *arXiv preprint:2109.12004*.

[Rockafellar, 1970] Rockafellar, R. (1970). *Convex Analysis*. Princeton Landmarks in Mathematics and Physics. Princeton University Press.

[Sadhu et al., 2024] Sadhu, R., Goldfeld, Z., and Kato, K. (2024). Stability and statistical inference for semidiscrete optimal transport maps. *The Annals of Applied Probability*, 34(6):5694–5736.

[Santambrogio, 2015] Santambrogio, F. (2015). Optimal transport for applied mathematicians. *Birkäuser, NY*, 55(58-63):94.

[Segers, 2022] Segers, J. (2022). Graphical and uniform consistency of estimated optimal transport plans. *arXiv preprint:2208.02508*.

[Seguy et al., 2018] Seguy, V., Damodaran, B. B., Flamary, R., Courty, N., Rolet, A., and Blondel, M. (2018). Large-scale optimal transport and mapping estimation. In *ICLR 2018-International Conference on Learning Representations*, pages 1–15.

[Staudt et al., 2025] Staudt, T., Hundrieser, S., and Munk, A. (2025). On the uniqueness of kantorovich potentials. *SIAM Journal on Mathematical Analysis*, 57(2):1452–1482.

[Vacher et al., 2021] Vacher, A., Muzellec, B., Rudi, A., Bach, F., and Vialard, F.-X. (2021). A dimension-free computational upper-bound for smooth optimal transport estimation. In *Conference on Learning Theory*, pages 4143–4173. PMLR.

[Vacher and Vialard, 2022] Vacher, A. and Vialard, F.-X. (2022). Parameter tuning and model selection in optimal transport with semi-dual Brenier formulation. *Advances in Neural Information Processing Systems*, 35:23098–23108.

[Villani, 2008] Villani, C. (2008). *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer Berlin Heidelberg.

[Villani, 2021] Villani, C. (2021). *Topics in optimal transportation*, volume 58. American Mathematical Soc.

[Weed, 2018] Weed, J. (2018). An explicit analysis of the entropic penalty in linear programming. In *Conference On Learning Theory*, pages 1841–1855. PMLR.